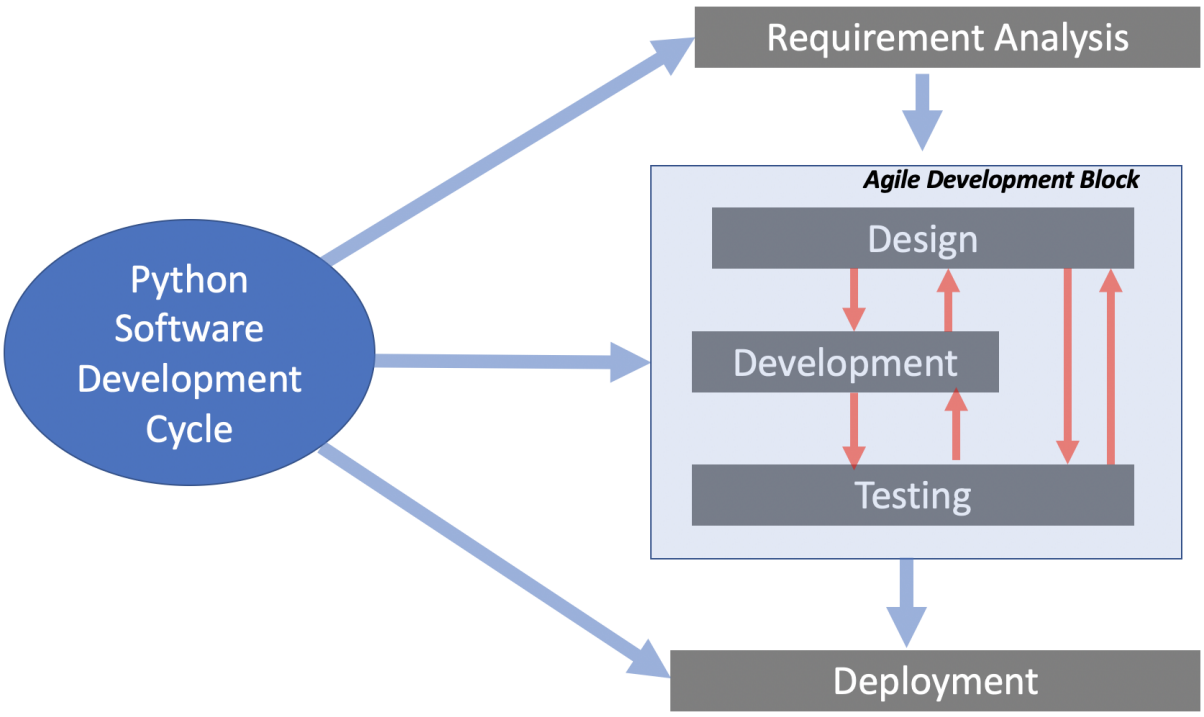
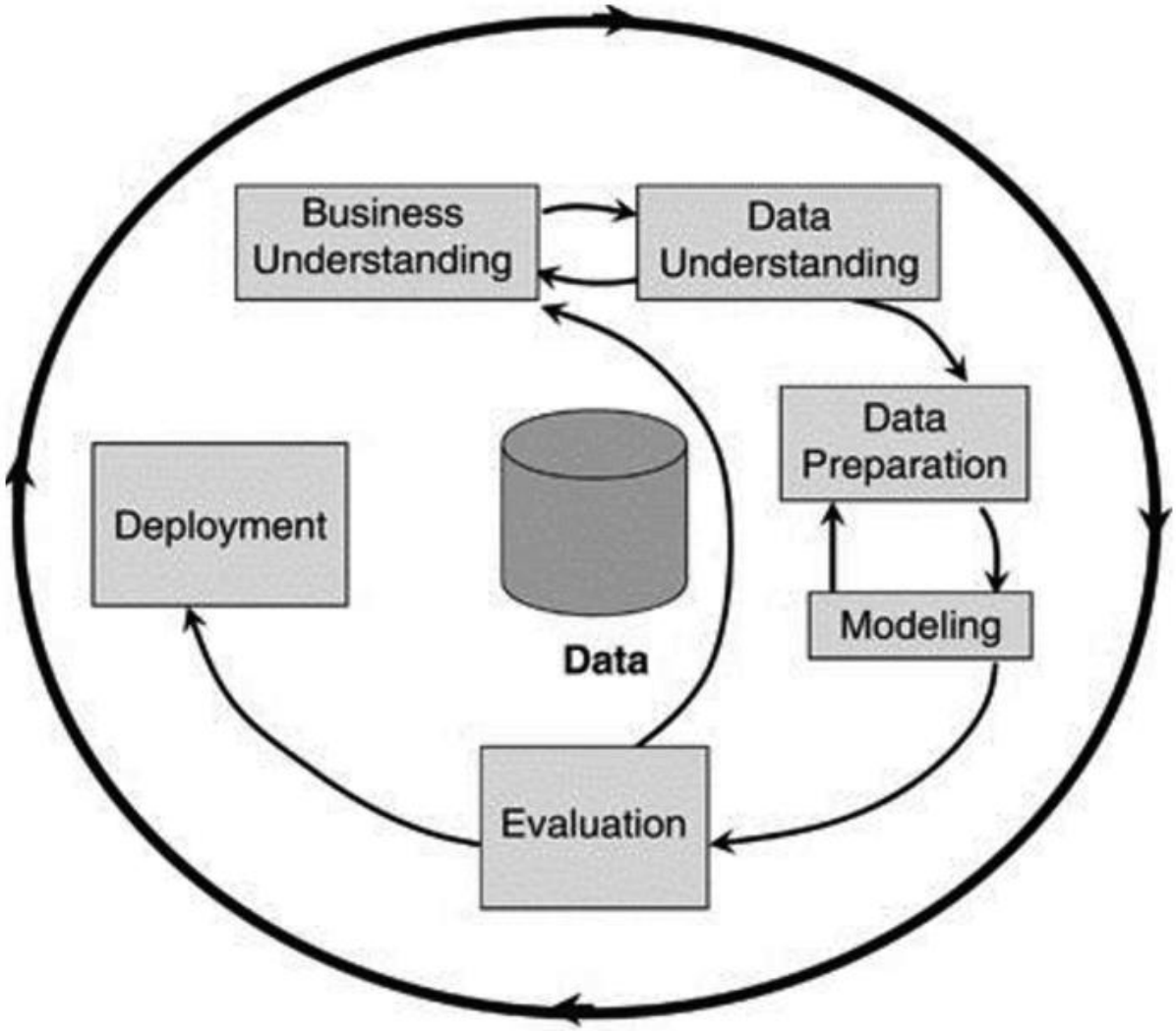
Chapter 1: Optimal Python Development Life Cycle

```
[1] import this
    this
```

```
The Zen of Python, by Tim Peters

Beautiful is better than ugly.
Explicit is better than implicit.
Simple is better than complex.
Complex is better than complicated.
Flat is better than nested.
Sparse is better than dense.
Readability counts.
Special cases aren't special enough to break the rules.
Although practicality beats purity.
Errors should never pass silently.
Unless explicitly silenced.
In the face of ambiguity, refuse the temptation to guess.
There should be one-- and preferably only one --obvious way to do it.
Although that way may not be obvious at first unless you're Dutch.
Now is better than never.
Although never is often better than *right* now.
If the implementation is hard to explain, it's a bad idea.
If the implementation is easy to explain, it may be a good idea.
Namespaces are one honking great idea -- let's do more of those!
<module 'this' from '/usr/lib/python3.6/this.py'>
```






```
# A simple example to demonstrate the comments.
print("This is Chapter 1")
```

```
This is Chapter 1
```

```
"""
This is a comment
written in
more than just one line
"""
print("Chapter 2 will be about Python Modules")
```

```
Chapter 2 will be about Python Modules
```

```
[5] def double(n):
        '''Takes in a number n, returns the double of its value'''
        return n**2
```

```
[6] print(double.__doc__)
```

```
Takes in a number n, returns the double of its value
```

```
[8] help(double)
```

```
Help on function double in module __main__:

double(n)
    Takes in a number n, returns the double of its value
```

```python
class ComplexNumber:
    """
    This is a class for mathematical operations on complex numbers.

    Attributes:
        real (int): The real part of complex number.
        imag (int): The imaginary part of complex number.
    """

    def __init__(self, real, imag):
        """
        The constructor for ComplexNumber class.

        Parameters:
           real (int): The real part of complex number.
           imag (int): The imaginary part of complex number.
        """

    def add(self, num):
        """
        The function to add two Complex Numbers.

        Parameters:
            num (ComplexNumber): The complex number to be added.

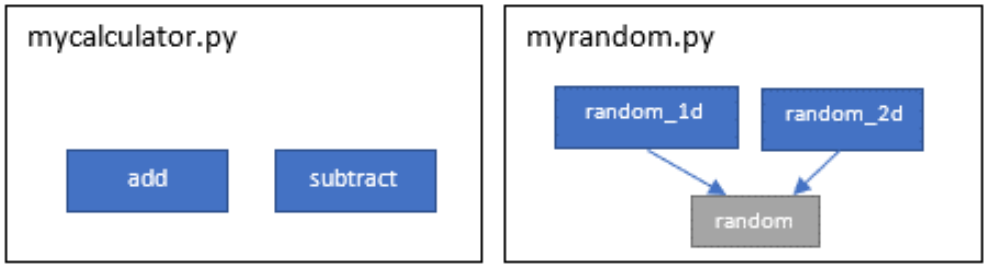
        Returns:
            ComplexNumber: A complex number which contains the sum.
        """

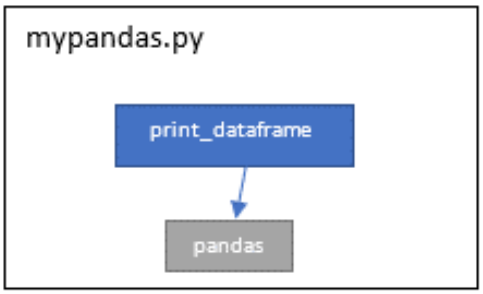
        re = self.real + num.real
        im = self.imag + num.imag

        return ComplexNumber(re, im)
```

Chapter 2: Using Modularization to Handle Complex Projects

Module Name	__name__ =
main.py	__main__
myrandom.py	myrandom
mycalculator.py	mycalculator

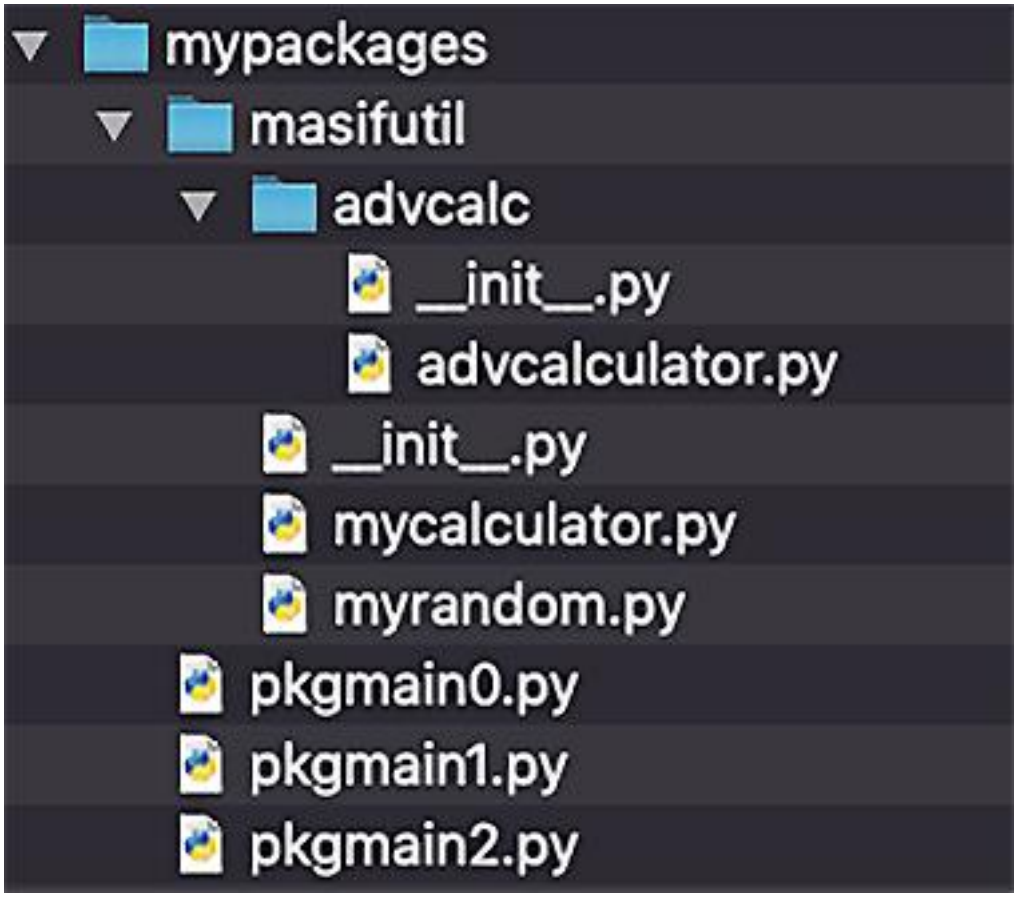


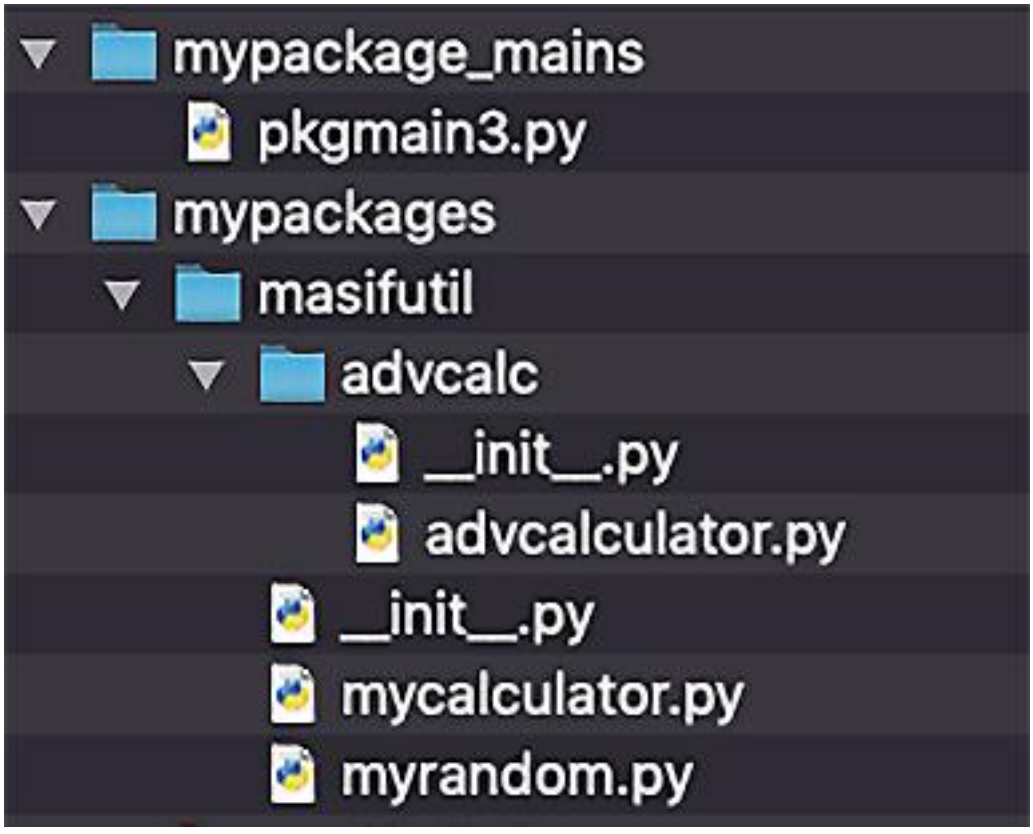


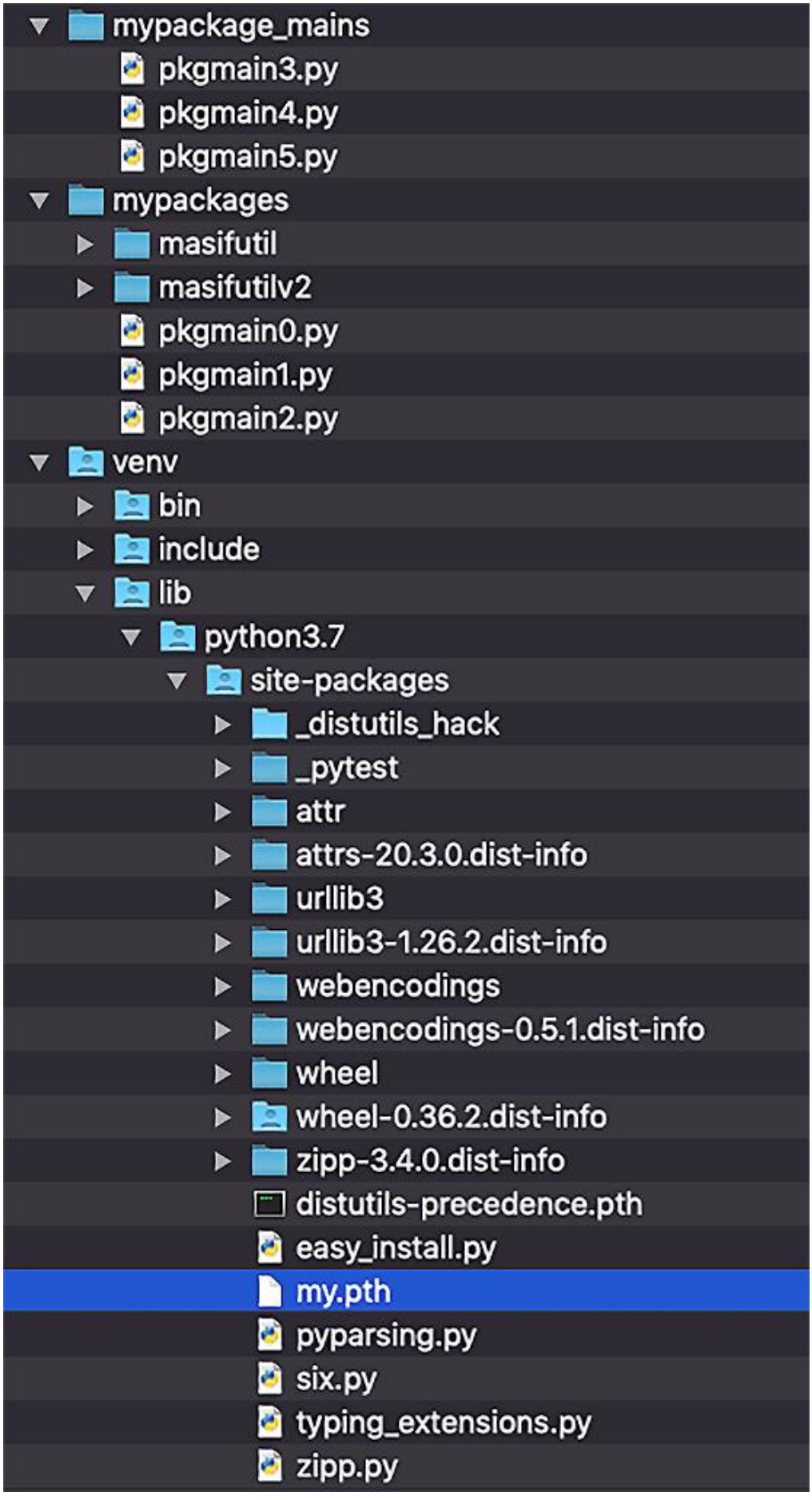

```
brics = pandas.tDataFrame(dict)
print(brics
```

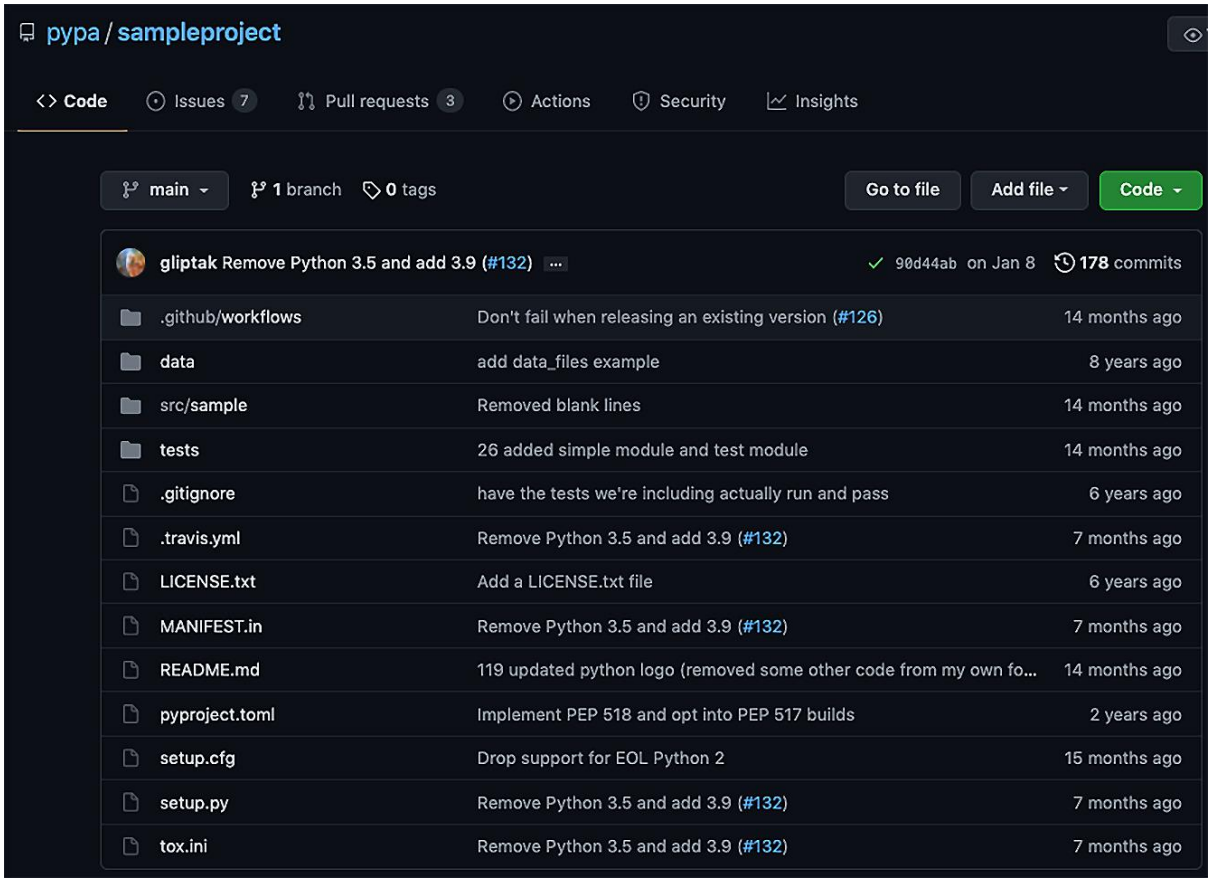
testing (pandas)
test(extra_args) pandas.util._tester
tseries (pandas)
timedelta_range(start, end, period… pandas.core.indexes.timedeltas
to_datetime(arg, errors, dayfirst, ye… pandas.core.tools.datetimes
to_numeric(arg, errors, downcast) pandas.core.tools.numeric
to_pickle(obj, filepath_or_buffer, compression, … pandas.io.pickle
to_timedelta(arg, unit, errors) pandas.core.tools.timedeltas
_tslib pandas
read_sql_table(table_name, con, schema, index_col, … pandas.io.sql
read_table pandas.io.parsers

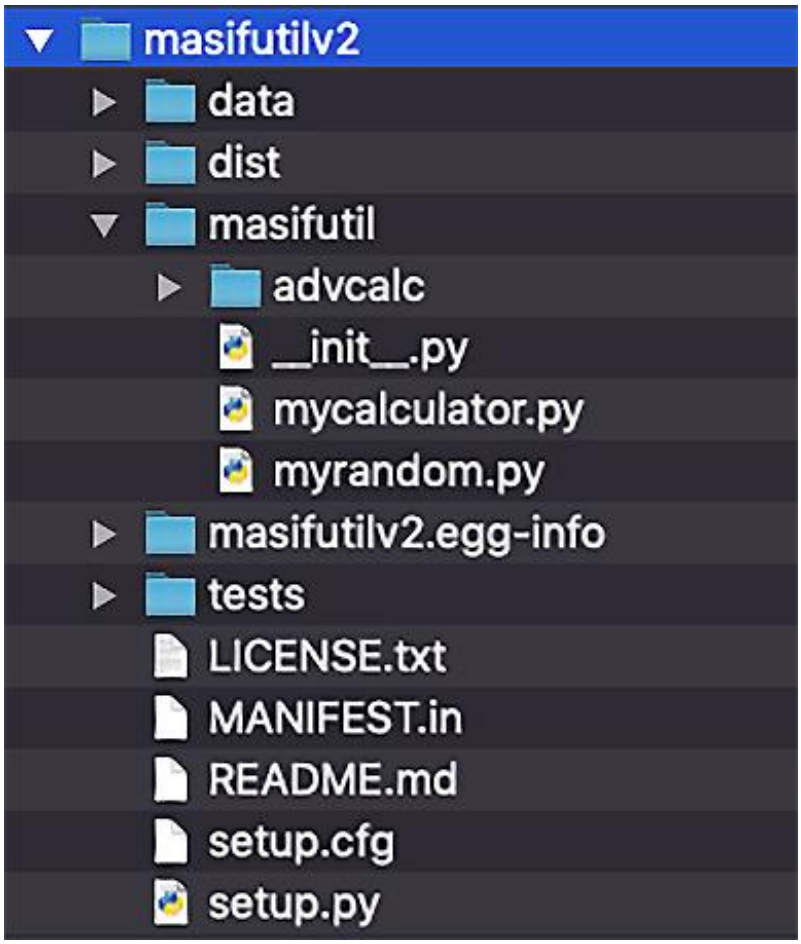
Press ^. to choose the selected (or first) suggestion and insert a dot afterwards Next Tip

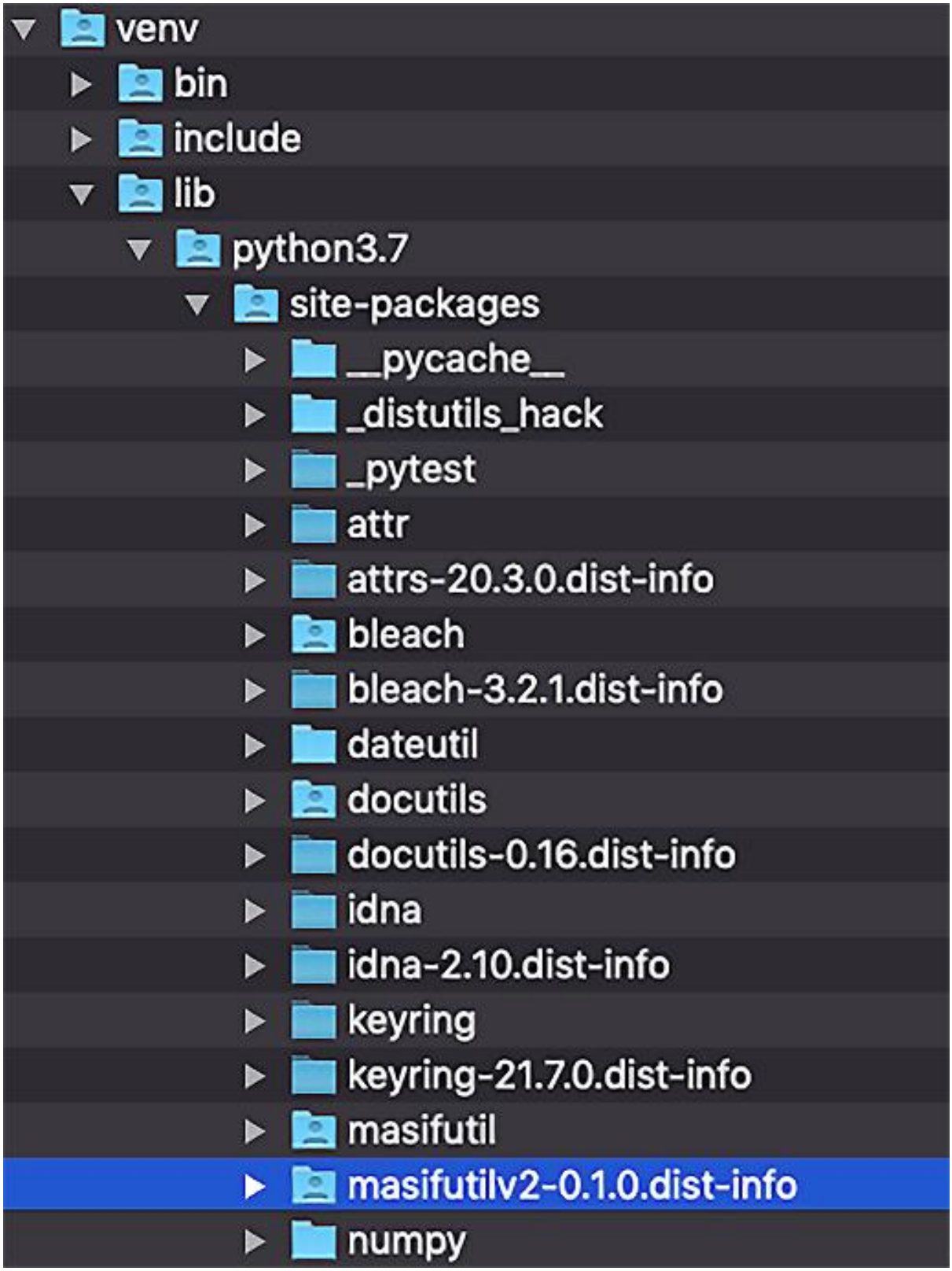

mypackages
masifutil
advcalc
__init__.py
advcalculator.py
__init__.py
mycalculator.py
myrandom.py
pkgmain0.py
pkgmain1.py
pkgmain2.py

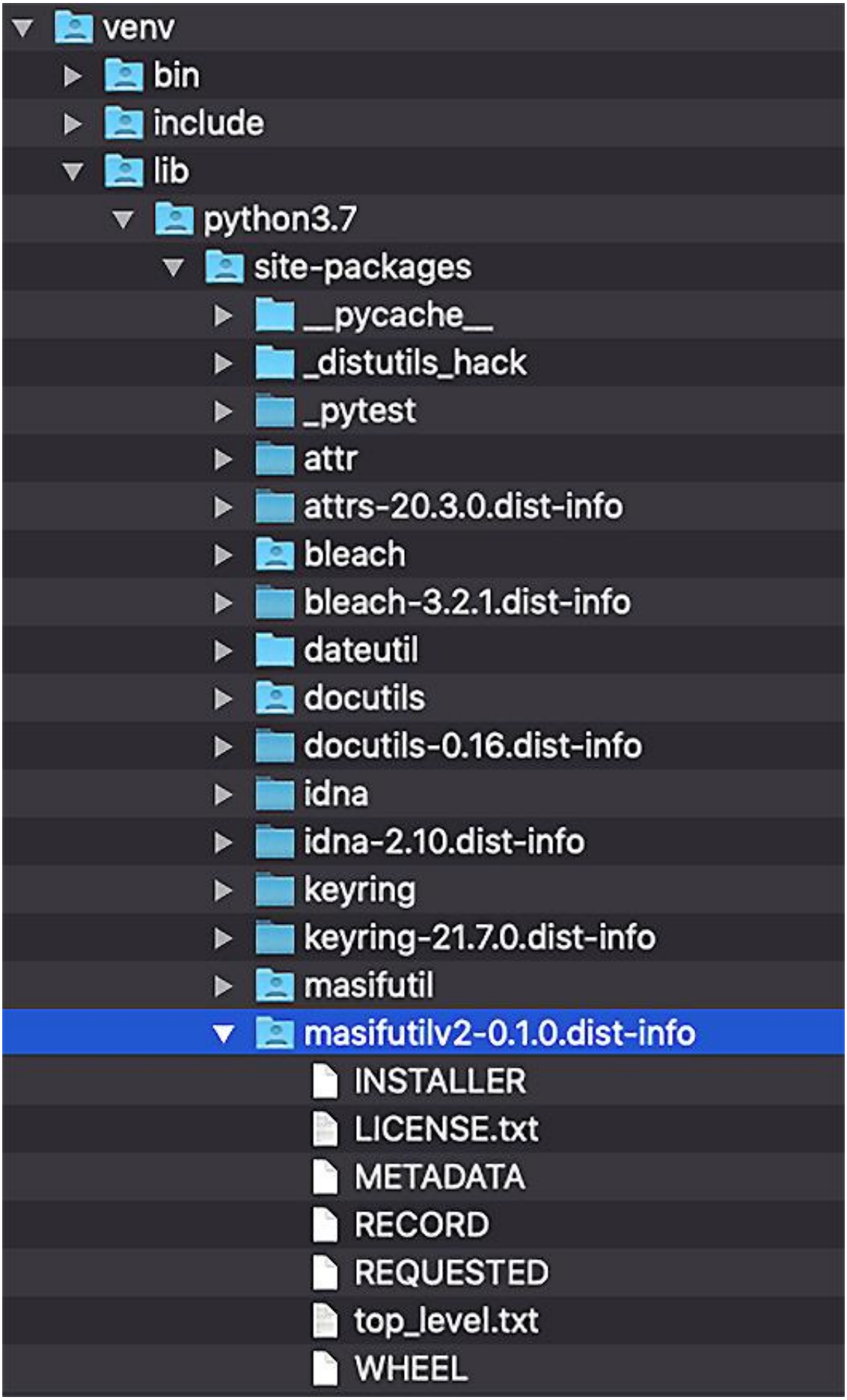

mypackage_mains
pkgmain3.py
mypackages
masifutil
advcalc
__init__.py
advcalculator.py
__init__.py
mycalculator.py
myrandom.py


mypackage_mains
pkgmain3.py
pkgmain4.py
pkgmain5.py
mypackages
masifutil
masifutilv2
pkgmain0.py
pkgmain1.py
pkgmain2.py
venv
bin
include
lib
python3.7
site-packages
_distutils_hack
_pytest
attr
attrs-20.3.0.dist-info
urllib3
urllib3-1.26.2.dist-info
webencodings
webencodings-0.5.1.dist-info
wheel
wheel-0.36.2.dist-info
zipp-3.4.0.dist-info
distutils-precedence.pth
easy_install.py
my.pth
pyparsing.py
six.py
typing_extensions.py
zipp.py

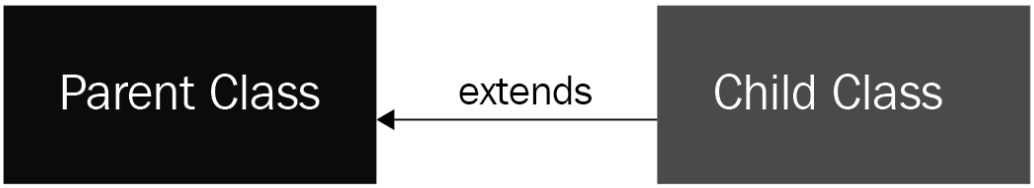

pypa / sampleproject
Code
Issues 7
Pull requests 3
Actions
Security
Insights
main
1 branch
0 tags
Go to file
Add file
Code
gliptak Remove Python 3.5 and add 3.9 (#132)
90d44ab on Jan 8
178 commits
.github/workflows
Don't fail when releasing an existing version (#126)
14 months ago
data
add data_files example
8 years ago
src/sample
Removed blank lines
14 months ago
tests
26 added simple module and test module
14 months ago
.gitignore
have the tests we're including actually run and pass
6 years ago
.travis.yml
Remove Python 3.5 and add 3.9 (#132)
7 months ago
LICENSE.txt
Add a LICENSE.txt file
6 years ago
MANIFEST.in
Remove Python 3.5 and add 3.9 (#132)
7 months ago
README.md
119 updated python logo (removed some other code from my own fo...
14 months ago
pyproject.toml
Implement PEP 518 and opt into PEP 517 builds
2 years ago
setup.cfg
Drop support for EOL Python 2
15 months ago
setup.py
Remove Python 3.5 and add 3.9 (#132)
7 months ago
tox.ini
Remove Python 3.5 and add 3.9 (#132)
7 months ago

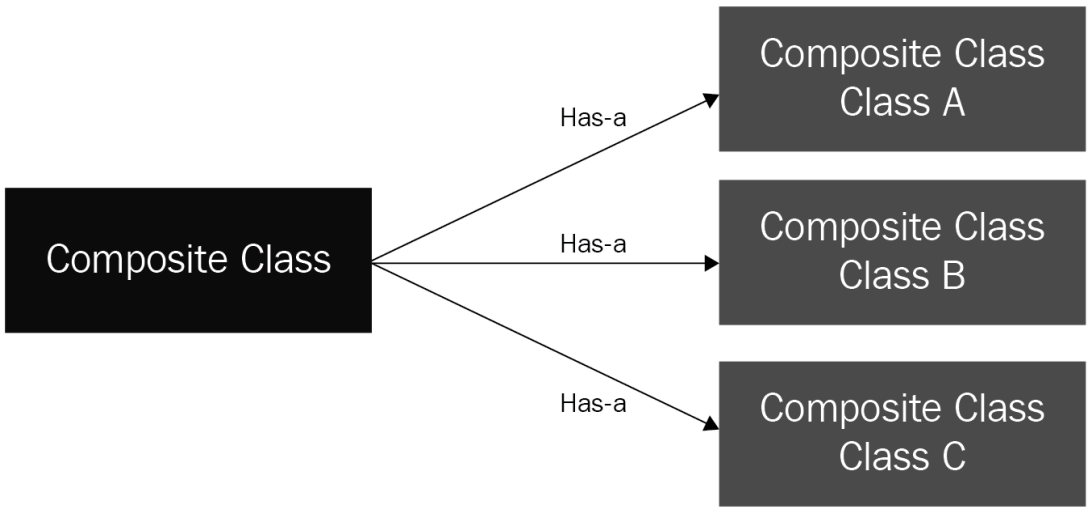

masifutilv2
data
dist
masifutil
advcalc
__init__.py
mycalculator.py
myrandom.py
masifutilv2.egg-info
tests
LICENSE.txt
MANIFEST.in
README.md
setup.cfg
setup.py


venv
bin
include
lib
python3.7
site-packages
__pycache__
_distutils_hack
_pytest
attr
attrs-20.3.0.dist-info
bleach
bleach-3.2.1.dist-info
dateutil
docutils
docutils-0.16.dist-info
idna
idna-2.10.dist-info
keyring
keyring-21.7.0.dist-info
masifutil
masifutilv2-0.1.0.dist-info
numpy

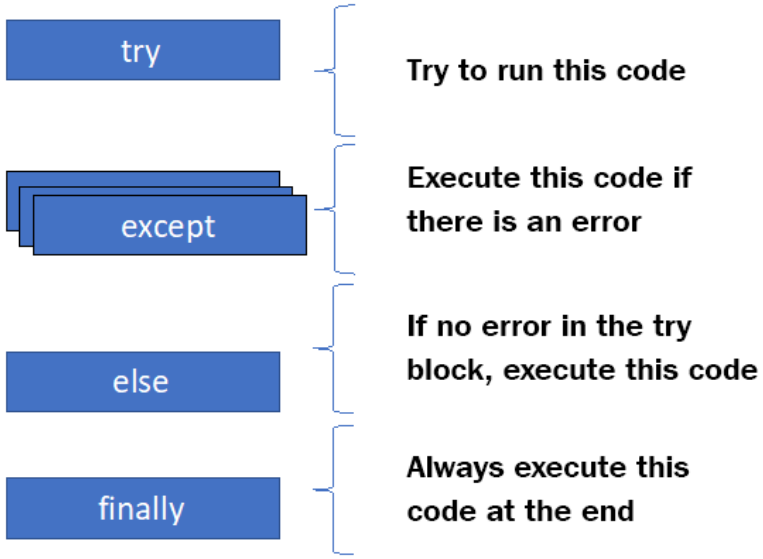

venv
bin
include
lib
python3.7
site-packages
__pycache__
_distutils_hack
_pytest
attr
attrs-20.3.0.dist-info
bleach
bleach-3.2.1.dist-info
dateutil
docutils
docutils-0.16.dist-info
idna
idna-2.10.dist-info
keyring
keyring-21.7.0.dist-info
masifutil
masifutilv2-0.1.0.dist-info
INSTALLER
LICENSE.txt
METADATA
RECORD
REQUESTED
top_level.txt
WHEEL

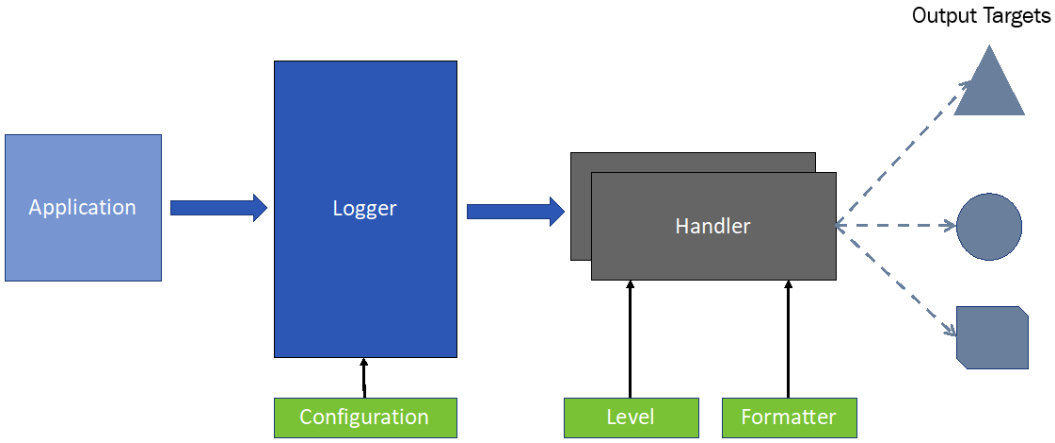
Chapter 3: Advanced Object-Oriented Python Programming

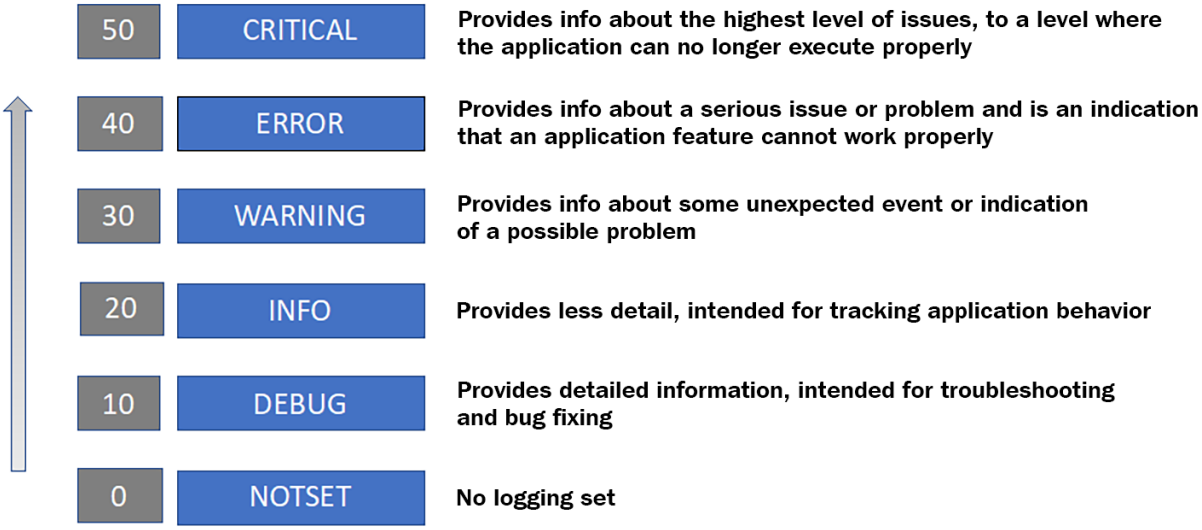




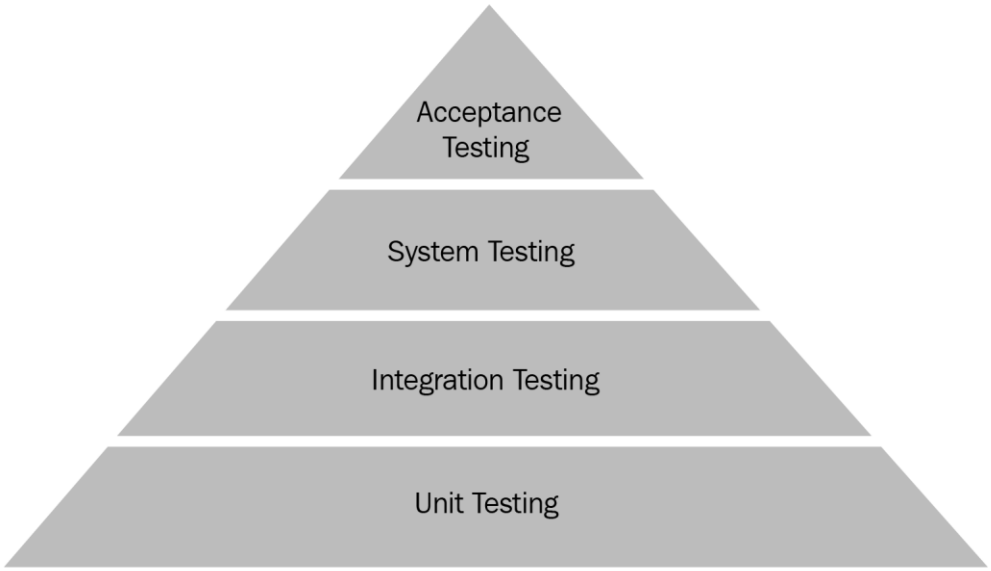
Chapter 4: Python Libraries for Advanced Programming



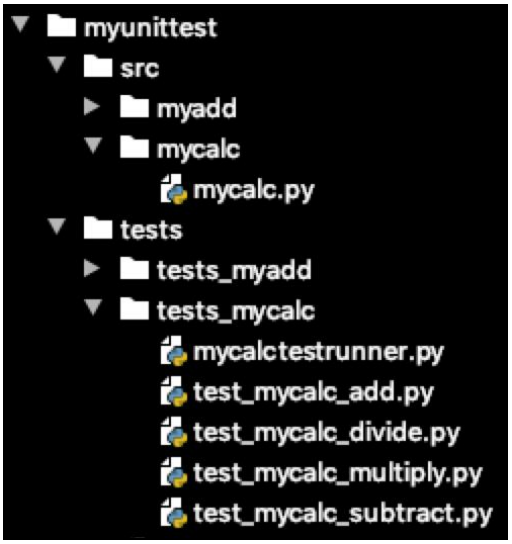


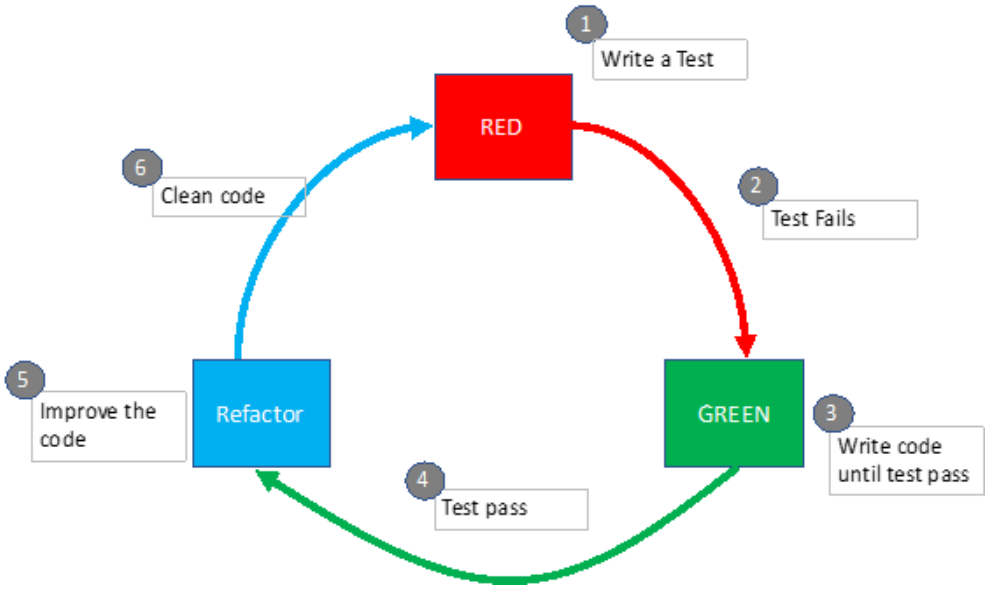


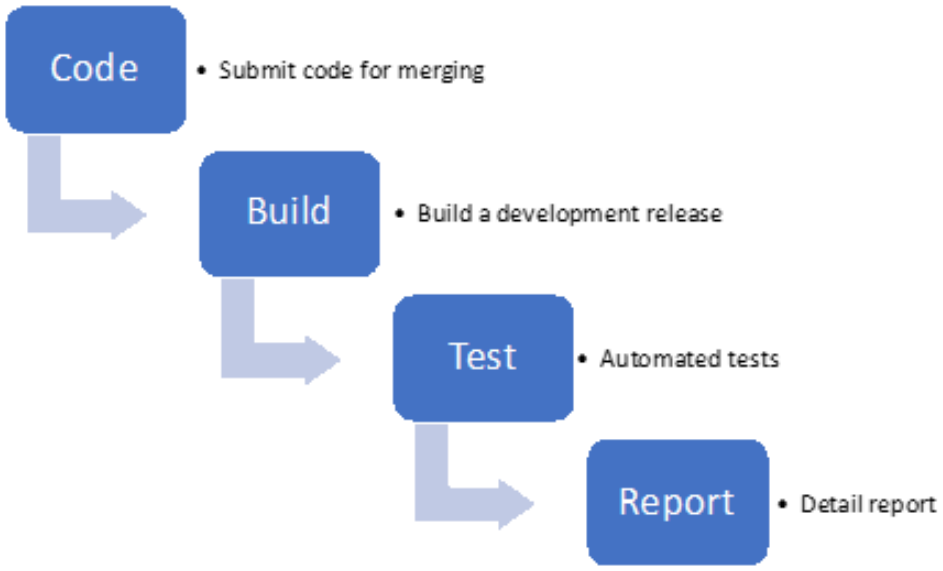
Chapter 5: Testing and Automation with Python



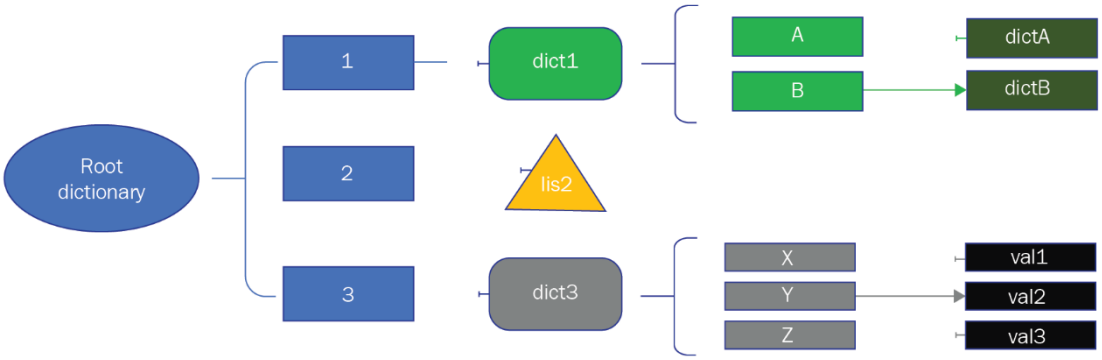
Method name	Evaluating condition
assertEqual (x, y)	Check if x is equal to y
assertTrue (x)	Check if x is Boolean true
assertFalse (x)	Check if x is Boolean false
assertNotEqual (x)	Check if x is not equal to y
assertIn (a, c)	Check if a is in collection c
assertNotIn (a, c)	Check if a is not in collection c
assertIs (x, y)	Check if x is y
assertIsNot (x, y)	Check if x is not y
assertIsNone (x)	Check if x is none


myunittest
src
myadd
mycalc
mycalc.py
tests
tests_myadd
tests_mycalc
mycalctestrunner.py
test_mycalc_add.py
test_mycalc_divide.py
test_mycalc_multiply.py
test_mycalc_subtract.py


1
Write a Test
RED
2
Test Fails
GREEN
3
Write code until test pass
4
Test pass
Refactor
5
Improve the code
6
Clean code


Code
• Submit code for merging
Build
• Build a development release
Test
• Automated tests
Report
• Detail report

Chapter 6: Advanced Tips and Tricks in Python

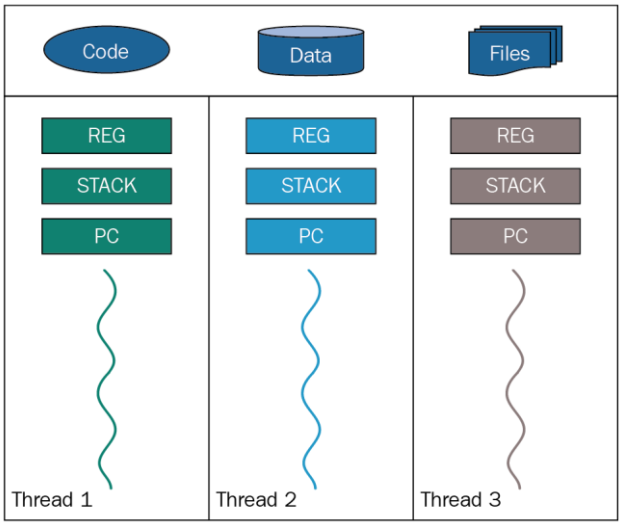


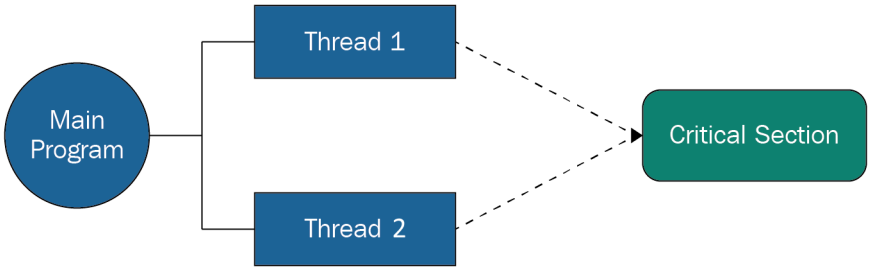
	day	temp	condition
0	Monday	40	Sunny
1	Tuesday	33	Cloudy
2	Wednesday	42	Sunny
3	Thursday	31	Rain
4	Friday	41	Sunny
5	Saturday	40	Cloudy
6	Sunday	30	Rain

day	temp	condition
Monday	40	Sunny
Tuesday	33	Cloudy
Wednesday	42	Sunny
Thursday	31	Rain
Friday	41	Sunny
Saturday	40	Cloudy
Sunday	30	Rain

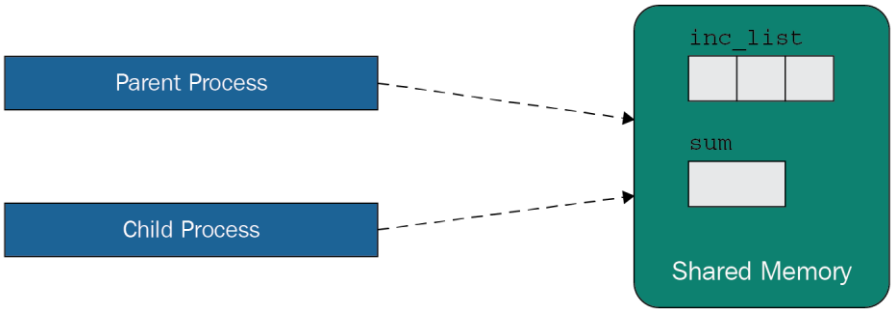
	day	temp	condition
MON	Monday	40	Sunny
TUE	Tuesday	33	Cloudy
WED	Wednesday	42	Sunny
THU	Thursday	31	Rain
FRI	Friday	41	Sunny
SAT	Saturday	40	Cloudy
SUN	Sunday	30	Rain

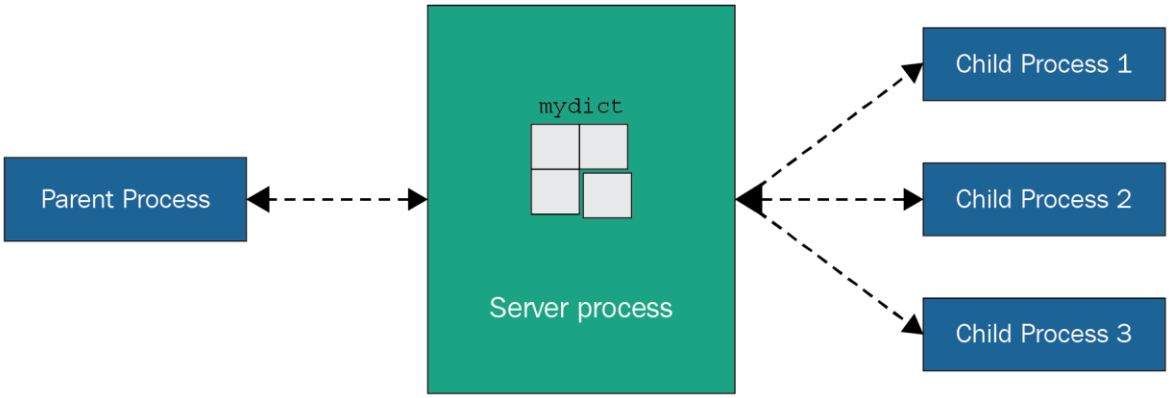
Chapter 7: Multiprocessing, Multithreading, and Asynchronous Programming



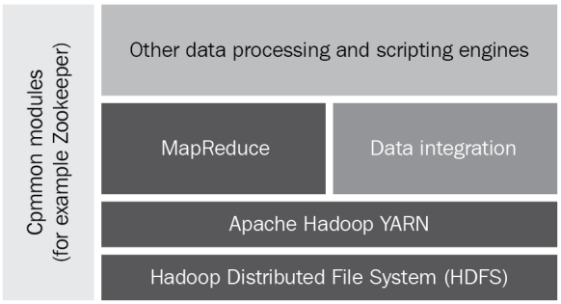


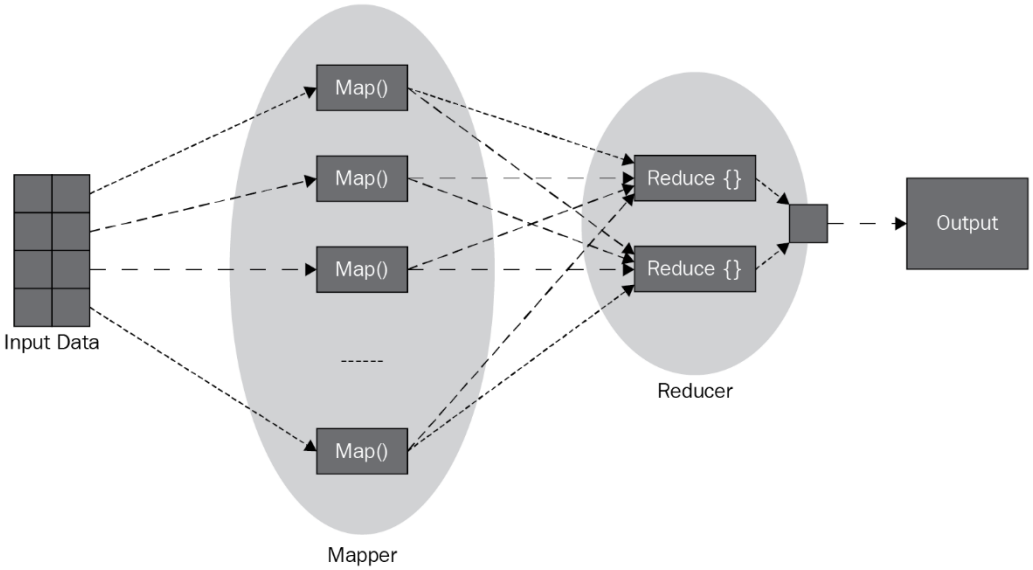
Using the Pool object	Using the Process object
Only active processes stay in memory	All created processes stay in memory
Works better for large datasets and for repetitive tasks	Works better for small datasets
Processes block on I/O operation until I/O resource is granted	Processes are not blocked on the I/O operation

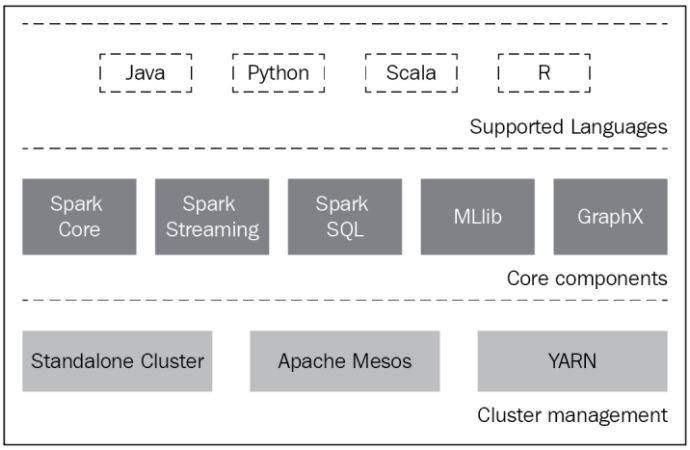


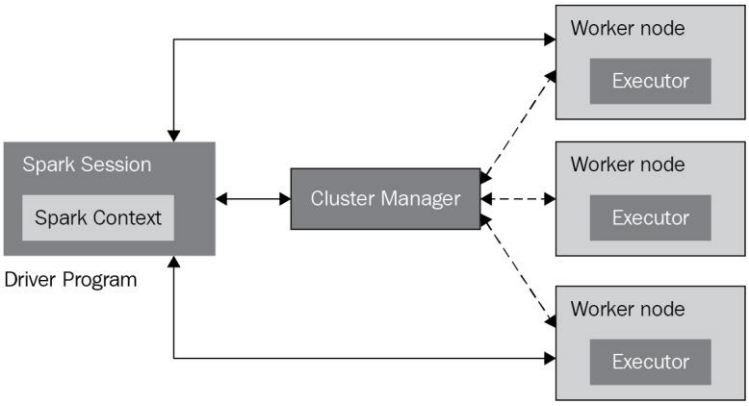

Parent Process
mydict
Server process
Child Process 1
Child Process 2
Child Process 3

Chapter 8: Scaling out Python Using Clusters









```
Welcome to
      ____              __
     / __/__  ___ _____/ /__
    _\ \/ _ \/ _ `/ __/  '_/
   /__ / .__/\_,_/_/ /_/\_\   version 3.1.1
      /_/

Using Python version 3.7.0 (v3.7.0:1bf9cc5093, Jun 26 2018 23:26:24)
Spark context Web UI available at http://192.168.1.110:4040
Spark context available as 'sc' (master = local[*], app id = local-1628712798632).
SparkSession available as 'spark'.
>>>
```

```
+---------+------+
|firstname|salary|
+---------+------+
|    James| 40000|
|    Kamal| 41000|
|   Robert| 35000|
|   Sophia| 47000|
|     John| 65000|
+---------+------+

+---------+----------+--------+-----------+------+------+---------+
|firstname|middlename|lastname| department|gender|salary|Pay Level|
+---------+----------+--------+-----------+------+------+---------+
|    James|          |  Bylsma|         HR|  Male| 40000|       11|
|    Kamal|     Rahim|        |         HR|  Male| 41000|       11|
|   Robert|          |   Zaine|    Finance|  Male| 35000|       10|
|   Sophia|      Anne|  Richer|    Finance|Female| 47000|       11|
|     John|      Will|   Brown|Engineering|Female| 65000|       12|
+---------+----------+--------+-----------+------+------+---------+
```

Name	State	IPv4	Image
vm1	Running	192.168.64.2	Ubuntu 20.04 LTS
vm2	Running	192.168.64.3	Ubuntu 20.04 LTS
vm3	Running	192.168.64.4	Ubuntu 20.04 LTS

Node Name	Role	Web UI
192.168.64.2	Master (Spark://192.168.64.2:7077)	http://192.168.64.2:8080/
192.168.64.3	Worker	http://192.168.64.3:8081/
192.168.64.4	Worker	http://192.168.64.4:8081/

Spark Master at spark://192.168.64.2:7077

URL: spark://192.168.64.2:7077
Alive Workers: 2
Cores in use: 2 Total, 0 Used
Memory in use: 2.0 GiB Total, 0.0 B Used
Resources in use:
Applications: 0 Running, 0 Completed
Drivers: 0 Running, 0 Completed
Status: ALIVE

Workers (2)

Worker Id	Address	State	Cores	Memory	Resources
worker-20210529145544-192.168.64.3-43027	192.168.64.3:43027	ALIVE	1 (0 Used)	1024.0 MiB (0.0 B Used)	
worker-20210529150201-192.168.64.4-34453	192.168.64.4:34453	ALIVE	1 (0 Used)	1024.0 MiB (0.0 B Used)	

Running Applications (0)

Completed Applications (0)

Spark Worker at 192.168.64.3:43027

ID: worker-20210529145544-192.168.64.3-43027
Master URL: spark://192.168.64.2:7077
Cores: 1 (0 Used)
Memory: 1024.0 MiB (0.0 B Used)
Resources:

Back to Master

Running Executors (0)

ExecutorID	State	Cores	Memory	Resources	Job Details	Logs

Finished Executors (13)

Spark Master at spark://192.168.64.2:7077

URL: spark://192.168.64.2:7077
Alive Workers: 2
Cores in use: 2 Total, 2 Used
Memory in use: 2.0 GiB Total, 2.0 GiB Used
Resources in use:
Applications: 1 Running, 0 Completed
Drivers: 0 Running, 0 Completed
Status: ALIVE

Workers (2)

Worker Id	Address	State	Cores	Memory	Resources
worker-20210529145544-192.168.64.3-43027	192.168.64.3:43027	ALIVE	1 (1 Used)	1024.0 MiB (1024.0 MiB Used)	
worker-20210529150201-192.168.64.4-34453	192.168.64.4:34453	ALIVE	1 (1 Used)	1024.0 MiB (1024.0 MiB Used)	

Running Applications (1)

Application ID	Name	Cores	Memory per Executor	Resources Per Executor	Submitted Time	User	State	Duration
app-20210529191451-0000 (kill)	Pi claculator app	2	1024.0 MiB		2021/05/29 19:14:51	muasif	RUNNING	9 s

Application: Pi claculator app

ID: app-20210529191451-0000
Name: Pi claculator app
User: muasif
Cores: Unlimited (2 granted)
Executor Limit: Unlimited (2 granted)
Executor Memory: 1024.0 MiB
Executor Resources:
Submit Date: 2021/05/29 19:14:51
State: FINISHED

Executor Summary (2)

ExecutorID	Worker	Cores	Memory	Resources	State	Logs

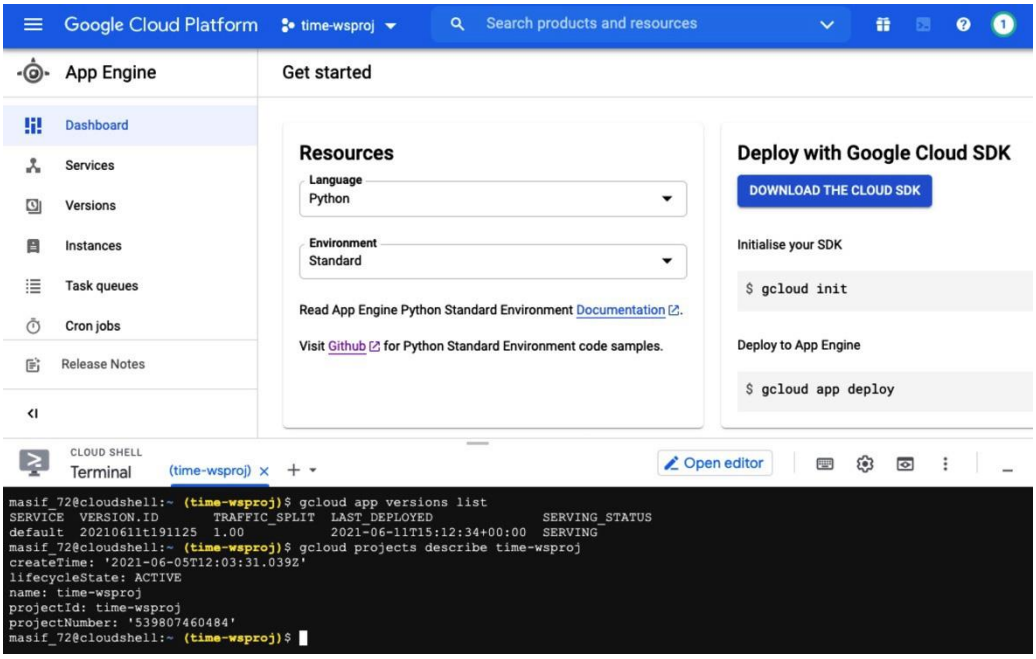
Removed Executors (2)

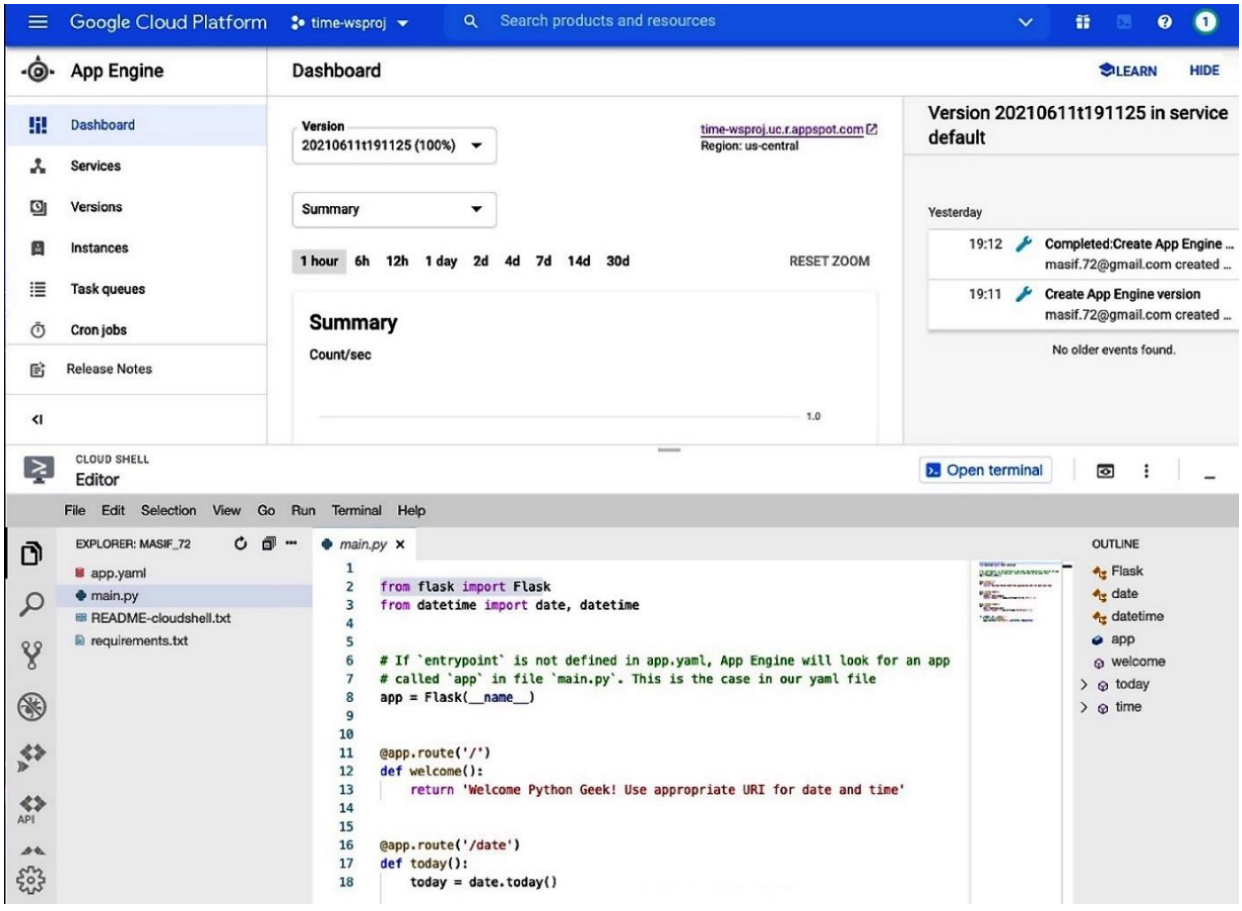
ExecutorID	Worker	Cores	Memory	Resources	State	Logs
1	worker-20210529150201-192.168.64.4-34453	1	1024		KILLED	stdout stderr
0	worker-20210529145544-192.168.64.3-43027	1	1024		KILLED	stdout stderr

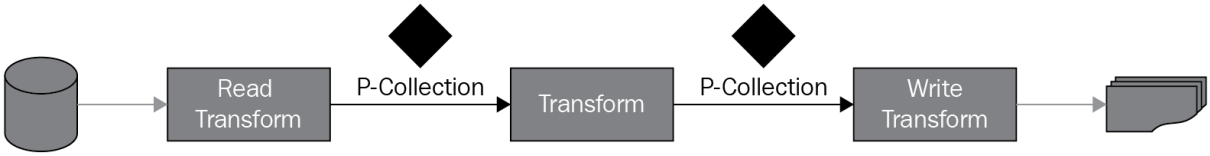


Chapter 9: Python Programming for the Cloud




Google Cloud Platform
time-wsproj
Search products and resources
App Engine
Get started
Dashboard
Services
Versions
Instances
Task queues
Cron jobs
Release Notes
Resources
Language
Python
Environment
Standard
Read App Engine Python Standard Environment Documentation.
Visit Github for Python Standard Environment code samples.
Deploy with Google Cloud SDK
DOWNLOAD THE CLOUD SDK
Initialise your SDK
$ gcloud init
Deploy to App Engine
$ gcloud app deploy
CLOUD SHELL
Terminal
(time-wsproj)
Open editor
masif_72@cloudshell:~ (time-wsproj)$ gcloud app versions list
SERVICE VERSION.ID TRAFFIC_SPLIT LAST_DEPLOYED SERVING_STATUS
default 20210611t191125 1.00 2021-06-11T15:12:34+00:00 SERVING
masif_72@cloudshell:~ (time-wsproj)$ gcloud projects describe time-wsproj
createTime: '2021-06-05T12:03:31.039Z'
lifecycleState: ACTIVE
name: time-wsproj
projectId: time-wsproj
projectNumber: '539807460484'
masif_72@cloudshell:~ (time-wsproj)$


Google Cloud Platform
time-wsproj
Search products and resources
App Engine
Dashboard
LEARN
HIDE
Dashboard
Services
Versions
Instances
Task queues
Cron jobs
Release Notes
Version
20210611t191125 (100%)
time-wsproj.uc.r.appspot.com
Region: us-central
Summary
1 hour 6h 12h 1 day 2d 4d 7d 14d 30d
RESET ZOOM
Summary
Count/sec
1.0
Version 20210611t191125 in service default
Yesterday
19:12 Completed:Create App Engine ...
masif.72@gmail.com created ...
19:11 Create App Engine version
masif.72@gmail.com created ...
No older events found.
CLOUD SHELL
Editor
Open terminal
File Edit Selection View Go Run Terminal Help
EXPLORER: MASIF_72
app.yaml
main.py
README-cloudshell.txt
requirements.txt
main.py
OUTLINE
Flask
date
datetime
app
welcome
today
time
API
1 from flask import Flask
2 from datetime import date, datetime
3
4
5 # If `entrypoint` is not defined in app.yaml, App Engine will look for an app
6 # called `app` in file `main.py`. This is the case in our yaml file
7 app = Flask(__name__)
8
9
10 @app.route('/')
11 def welcome():
12 return 'Welcome Python Geek! Use appropriate URI for date and time'
13
14
15 @app.route('/date')
16 def today():
17 today = date.today()
18

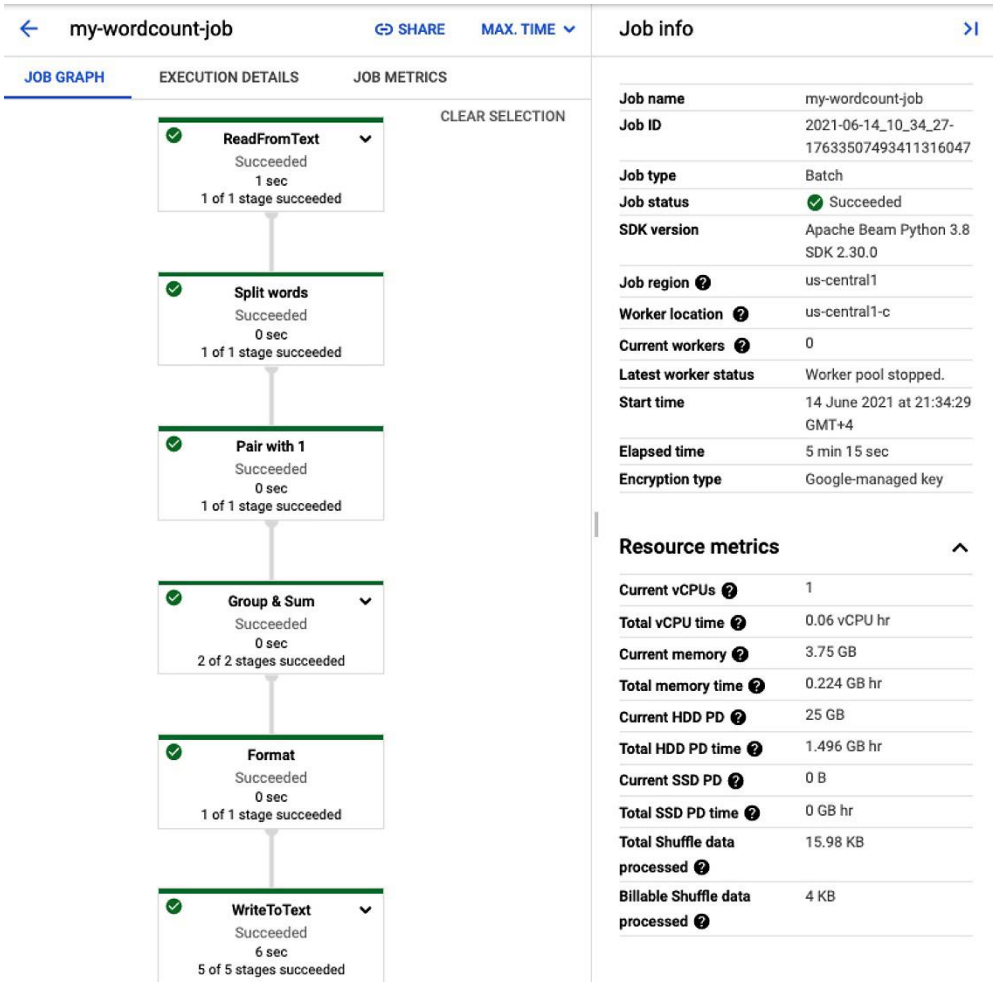

Read Transform
P-Collection
Transform
P-Collection
Write Transform

Jobs

CREATE JOB FROM TEMPLATE | CREATE JOB FROM SQL | ENABLE SORTING | REFRESH | LEARN

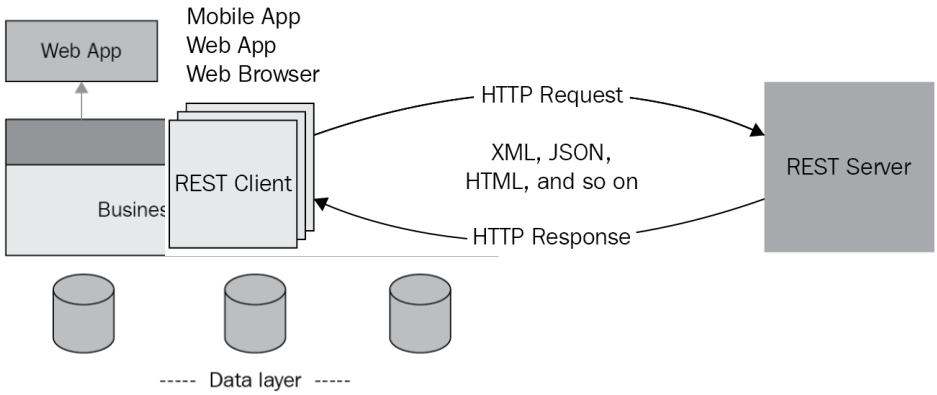
Running | Filter | Filter jobs

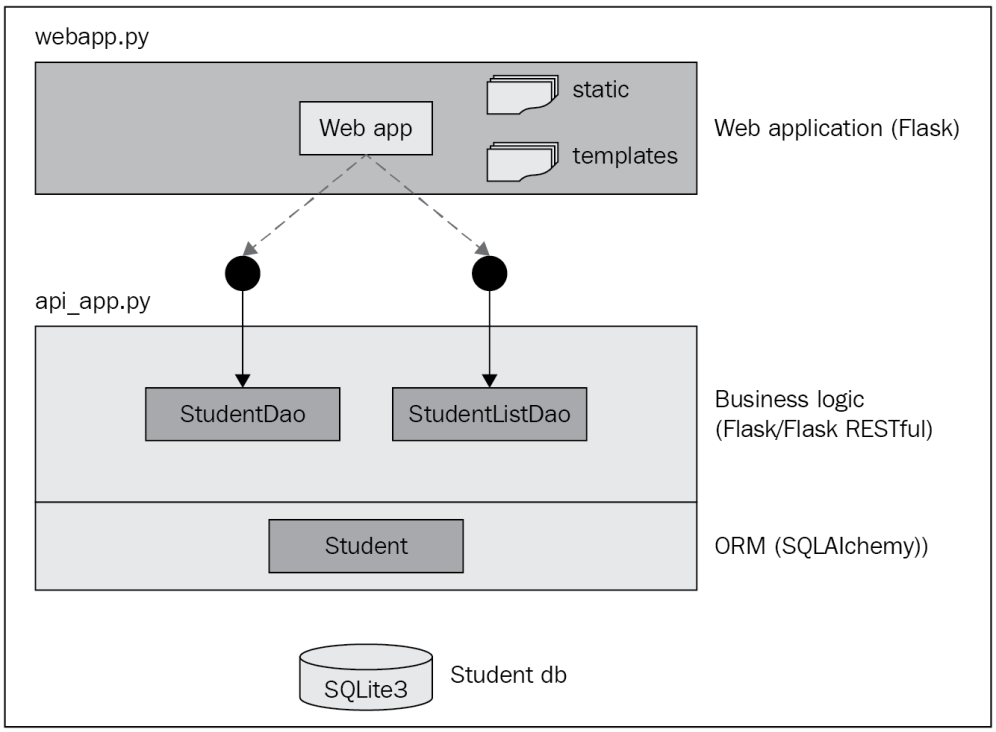
Name	Type	End time	Elapsed time	Start time	Status	SDK version	ID	Region
my-wordcount-job	Batch	14 Jun 2021, 21:39:43	5 min 14 sec	14 Jun 2021, 21:34:29	Succeeded	2.30.0	2021-06-14_10_34_27-17633507493411316047	us-central1
my-wordcount-job	Batch	14 Jun 2021, 21:30:54	5 min 3 sec	14 Jun 2021, 21:25:51	Succeeded	2.30.0	2021-06-14_10_25_49-9276325152354335464	us-central1
my-wordcount-job	Batch	14 Jun 2021, 13:56:34	4 min 43 sec	14 Jun 2021, 13:51:51	Failed	2.30.0	2021-06-14_02_51_50-7355109158749375121	us-central1

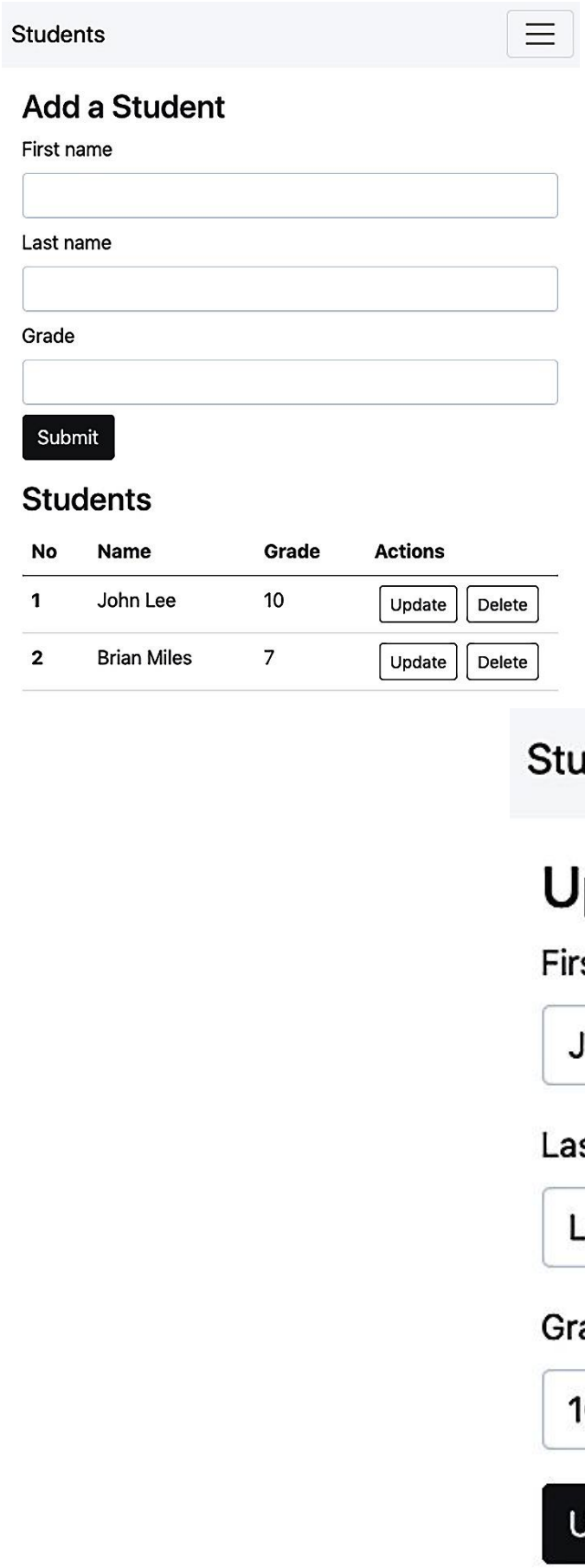


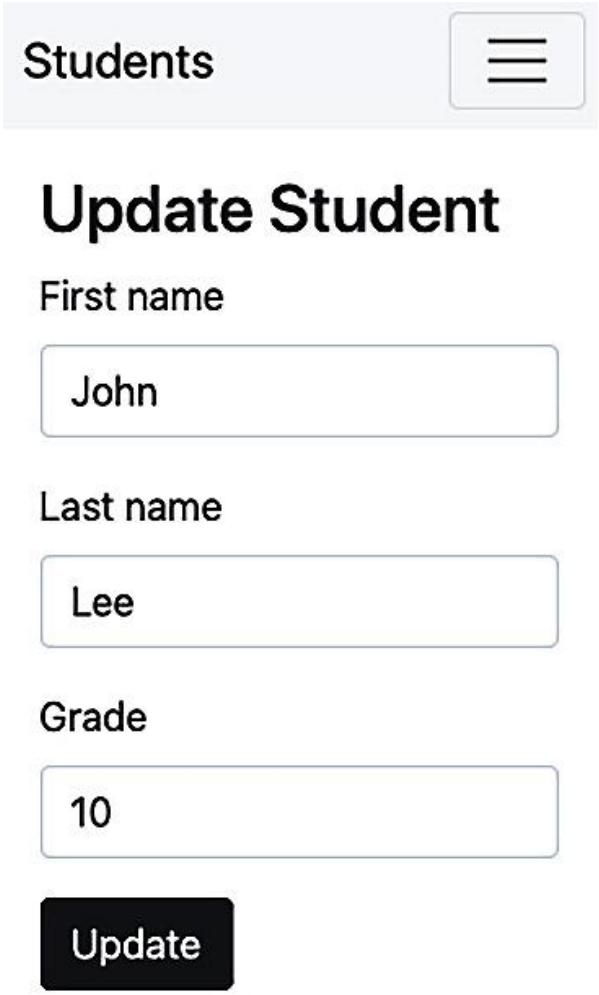
Chapter 10: Using Python for Web Development and REST API

Error Code	Name	Description
400	Bad Request	This error indicates either the URL or the request contents are incorrect.
401	Unauthorized	This error appears when the username or password is not provided or not correctly provided with the request.
403	Forbidden	This error shows that a user is trying to access a resource that is not allowed for that user.
404	Not Found	This error is returned when we are trying to access a resource that is not available. This is usually for the wrong URL.
500	Internal Server Error	This error indicates that the request is correct but that something happened on the server side.

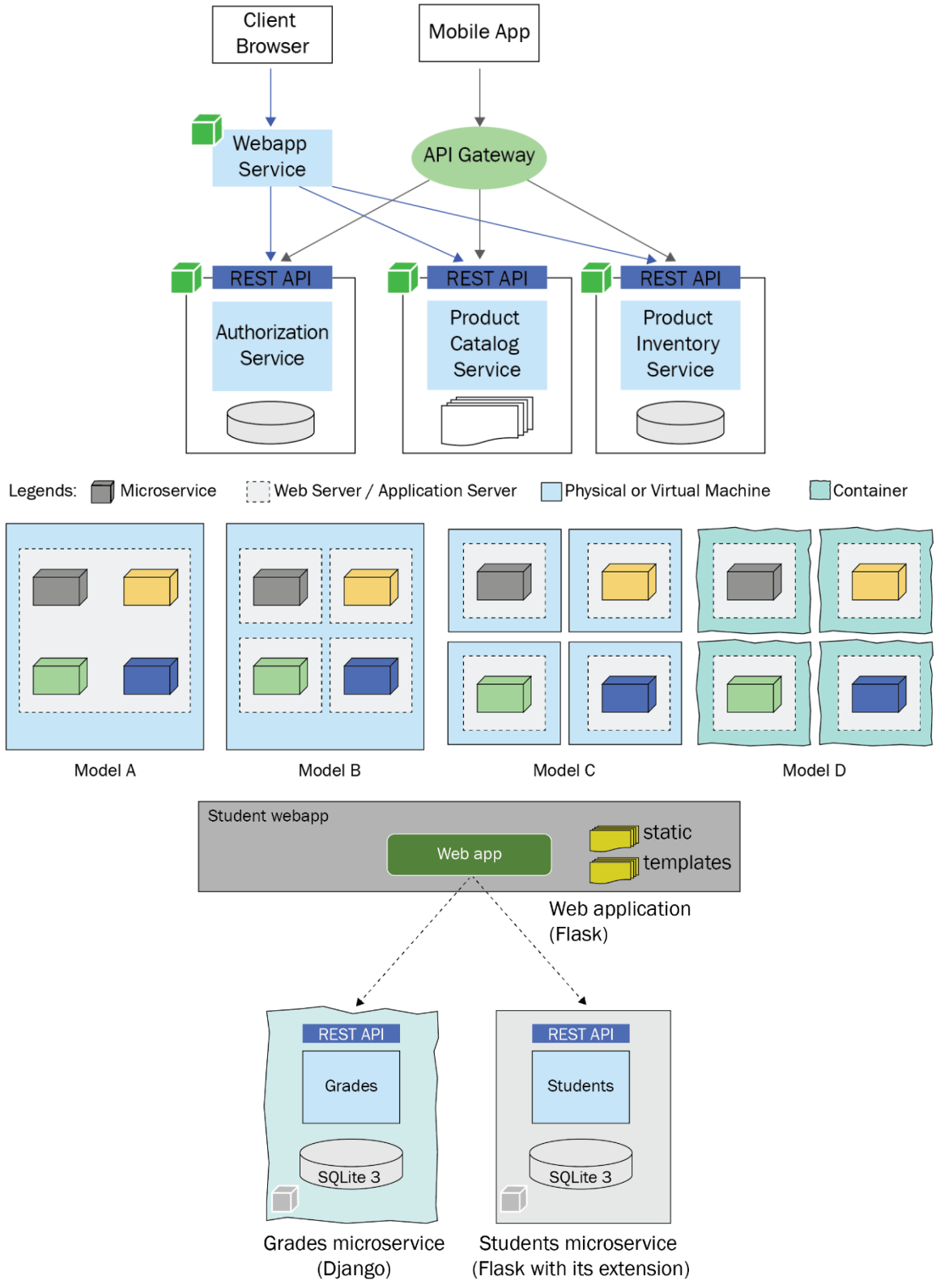


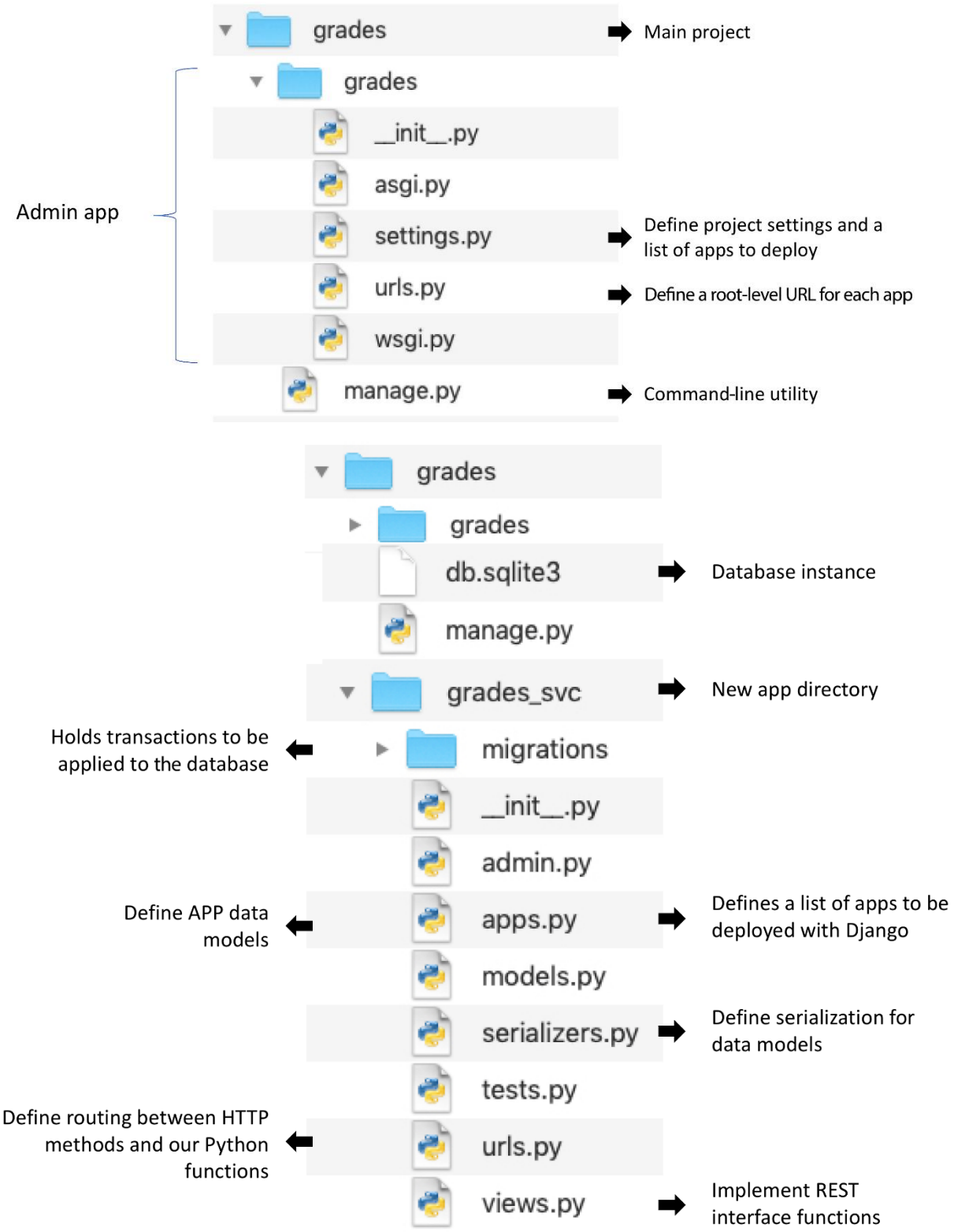


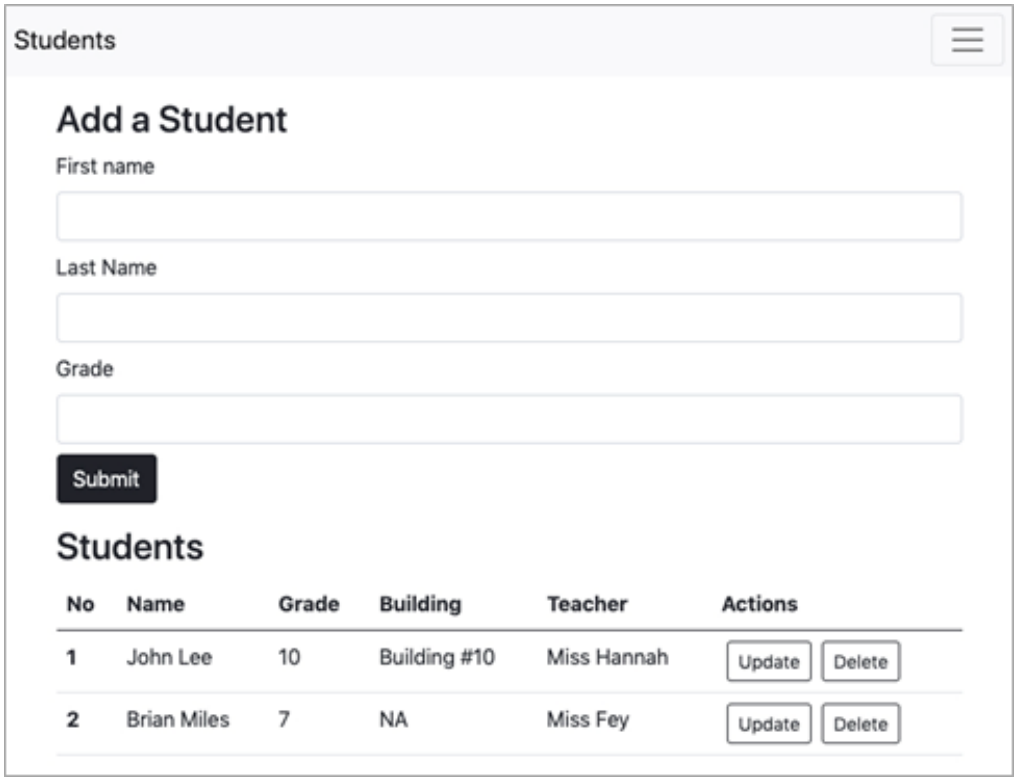

Students
Add a Student
First name
Last name
Grade
Submit
Students
No
Name
Grade
Actions
1
John Lee
10
Update
Delete
2
Brian Miles
7
Update
Delete


Students
Update Student
First name
John
Last name
Lee
Grade
10
Update

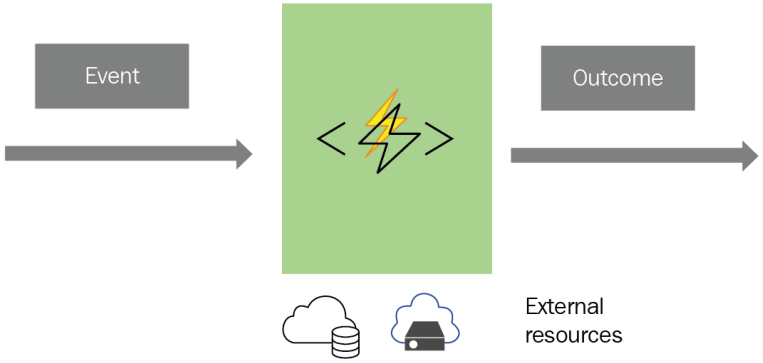
Chapter 11: Using Python for Microservices Development

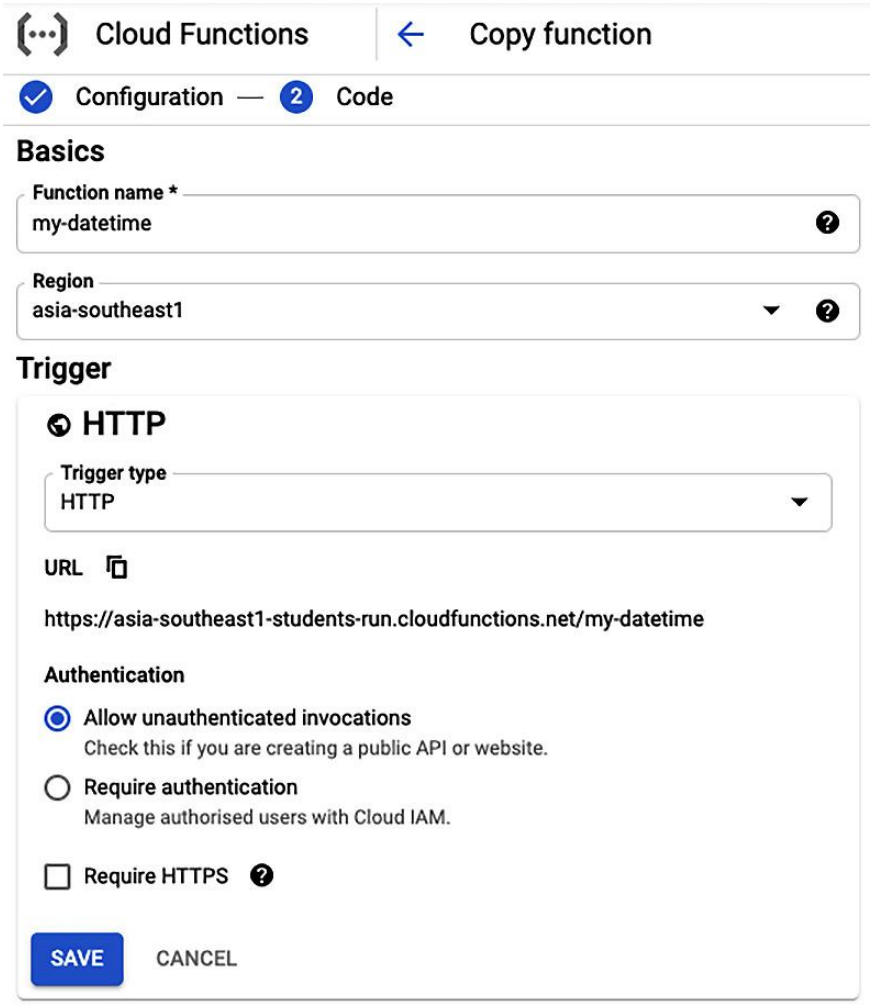


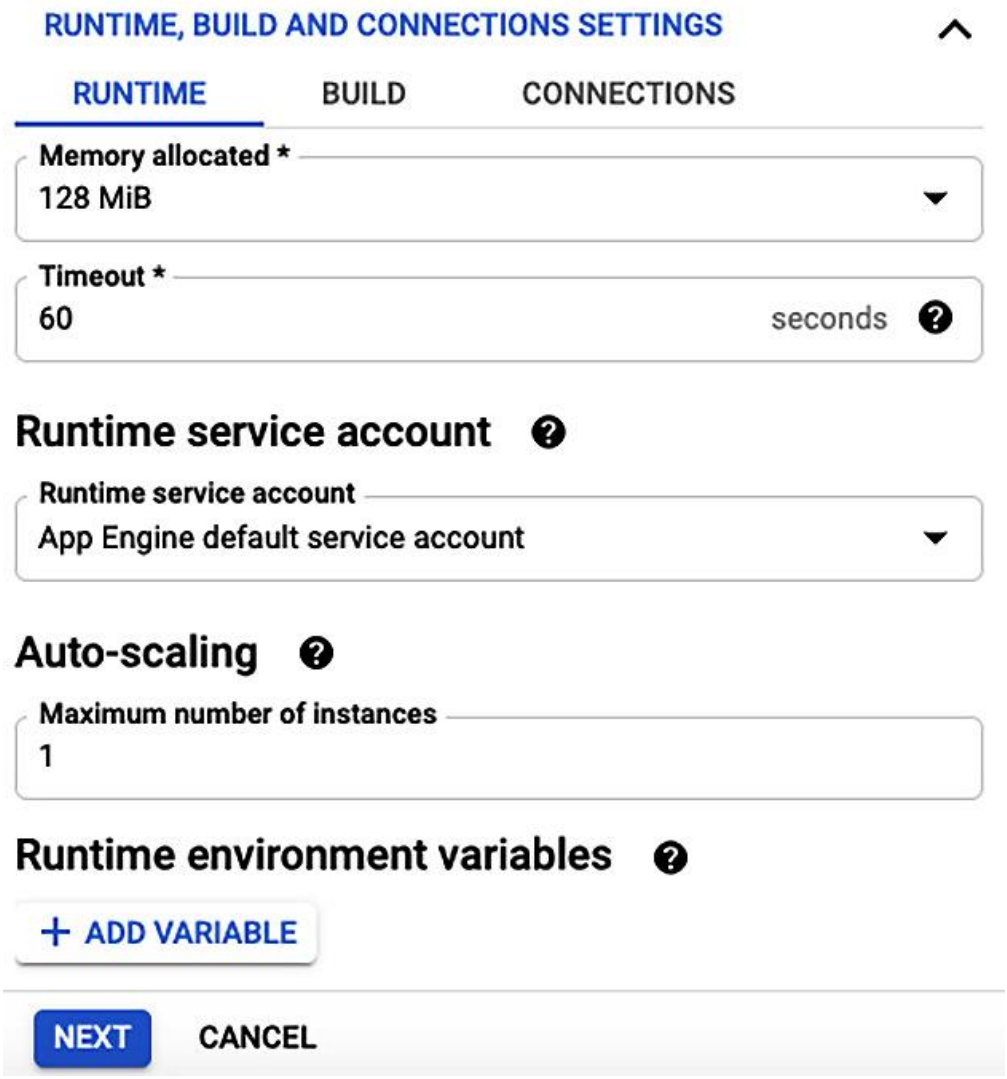

grades
Main project
grades
__init__.py
asgi.py
Admin app
settings.py
Define project settings and a list of apps to deploy
urls.py
Define a root-level URL for each app
wsgi.py
manage.py
Command-line utility
grades
grades
db.sqlite3
Database instance
manage.py
grades_svc
New app directory
Holds transactions to be applied to the database
migrations
__init__.py
admin.py
Define APP data models
apps.py
Defines a list of apps to be deployed with Django
models.py
serializers.py
Define serialization for data models
tests.py
Define routing between HTTP methods and our Python functions
urls.py
views.py
Implement REST interface functions

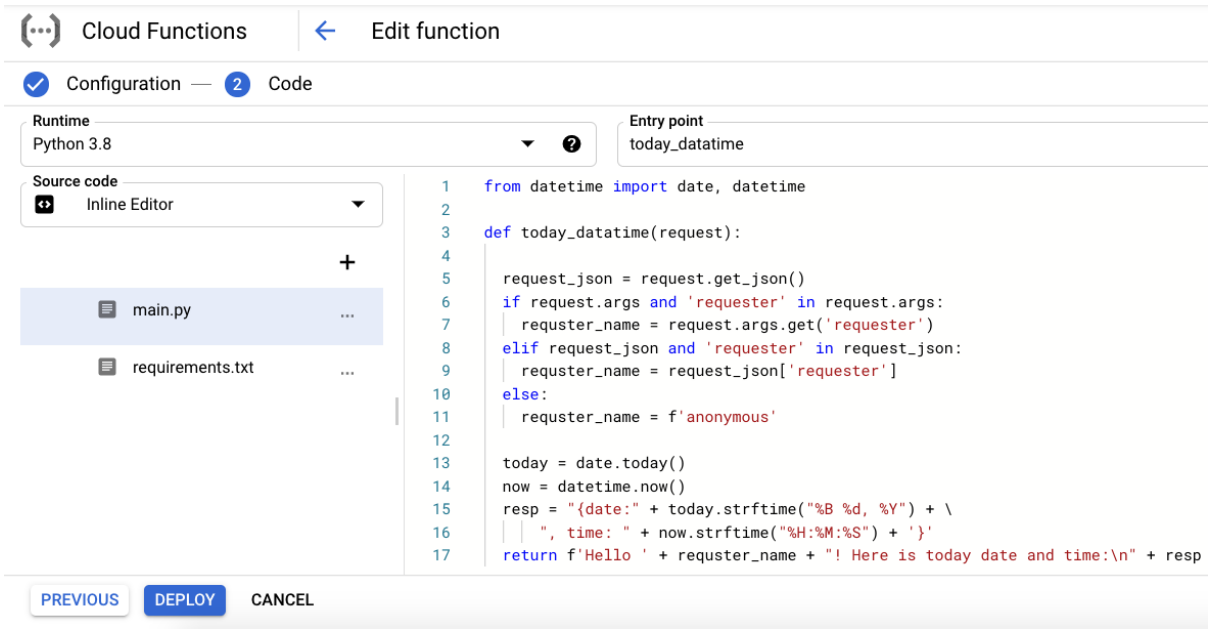

Students
Add a Student
First name
Last Name
Grade
Submit
Students
No
Name
Grade
Building
Teacher
Actions
1
John Lee
10
Building #10
Miss Hannah
Update
Delete
2
Brian Miles
7
NA
Miss Fey
Update
Delete

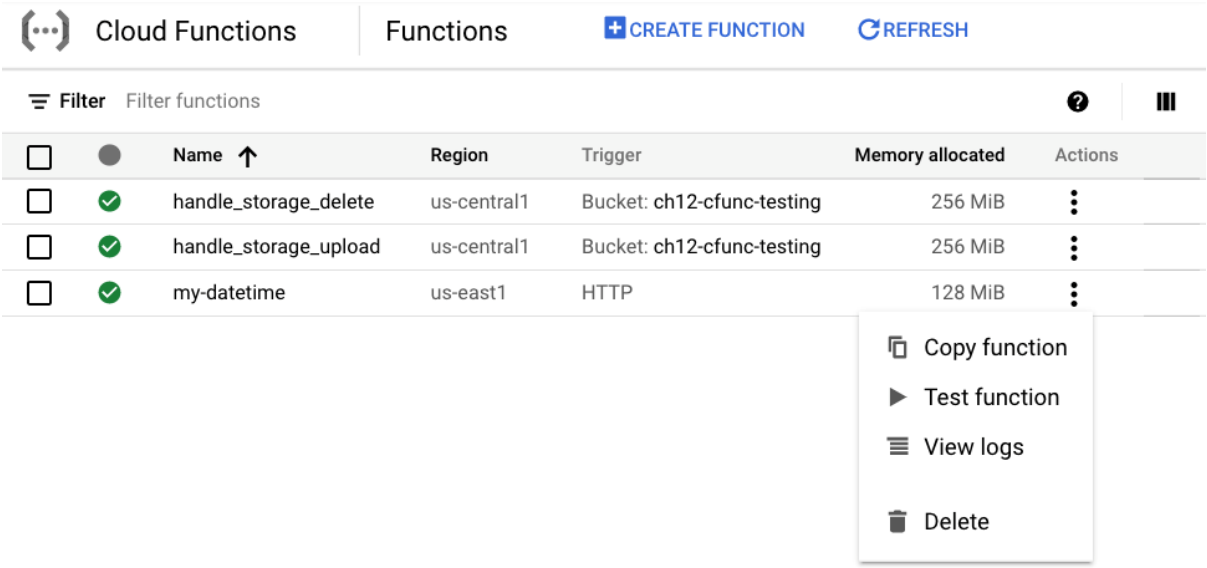
Chapter 12: Building Serverless Functions Using Python

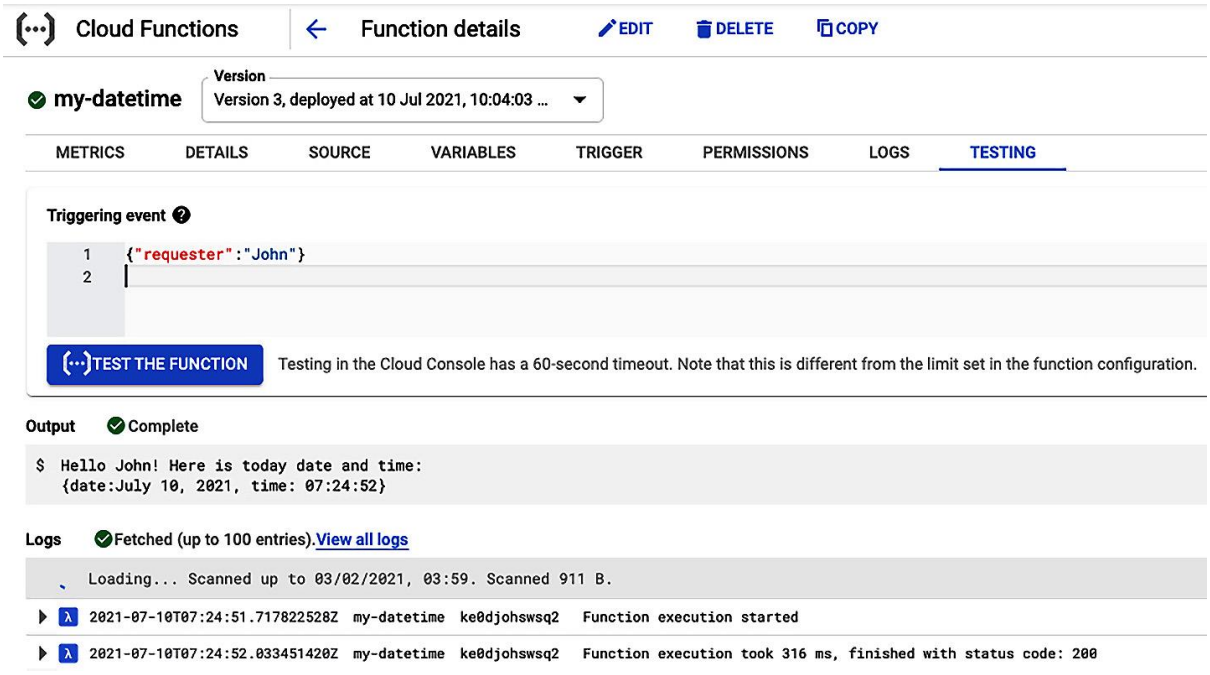


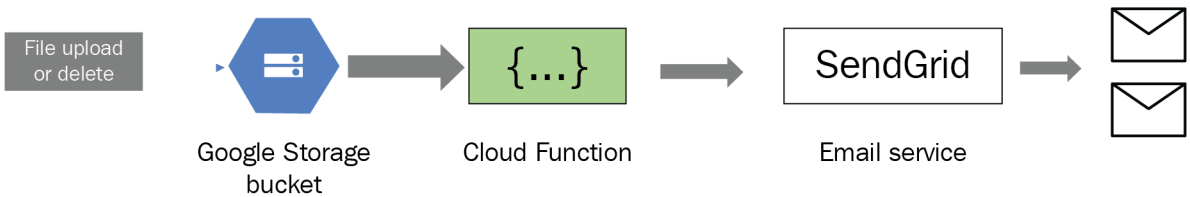



RUNTIME, BUILD AND CONNECTIONS SETTINGS
RUNTIME
BUILD
CONNECTIONS
Memory allocated *
128 MiB
Timeout *
60
seconds
Runtime service account
Runtime service account
App Engine default service account
Auto-scaling
Maximum number of instances
1
Runtime environment variables
ADD VARIABLE
NEXT
CANCEL


Cloud Functions
Edit function
Configuration
2
Code
Runtime
Python 3.8
Entry point
today_datatime
Source code
Inline Editor
main.py
requirements.txt
1 from datetime import date, datetime
2
3 def today_datatime(request):
4
5 request_json = request.get_json()
6 if request.args and 'requester' in request.args:
7 requster_name = request.args.get('requester')
8 elif request_json and 'requester' in request_json:
9 requster_name = request_json['requester']
10 else:
11 requster_name = f'anonymous'
12
13 today = date.today()
14 now = datetime.now()
15 resp = "{date:" + today.strftime("%B %d, %Y") + \
16 ", time: " + now.strftime("%H:%M:%S") + '}'
17 return f'Hello ' + requster_name + "! Here is today date and time:\n" + resp
PREVIOUS
DEPLOY
CANCEL





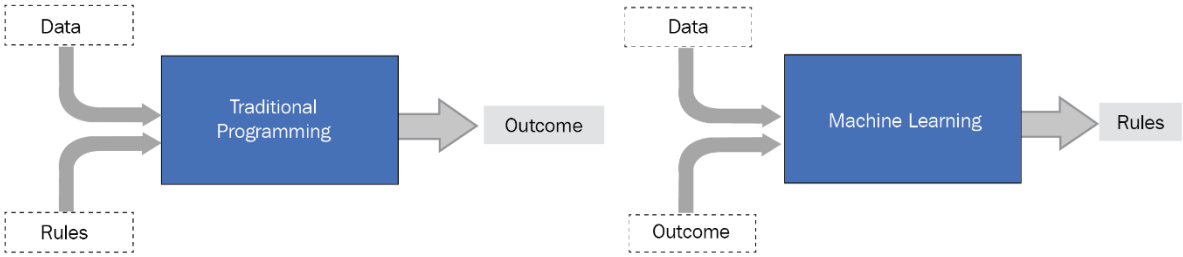


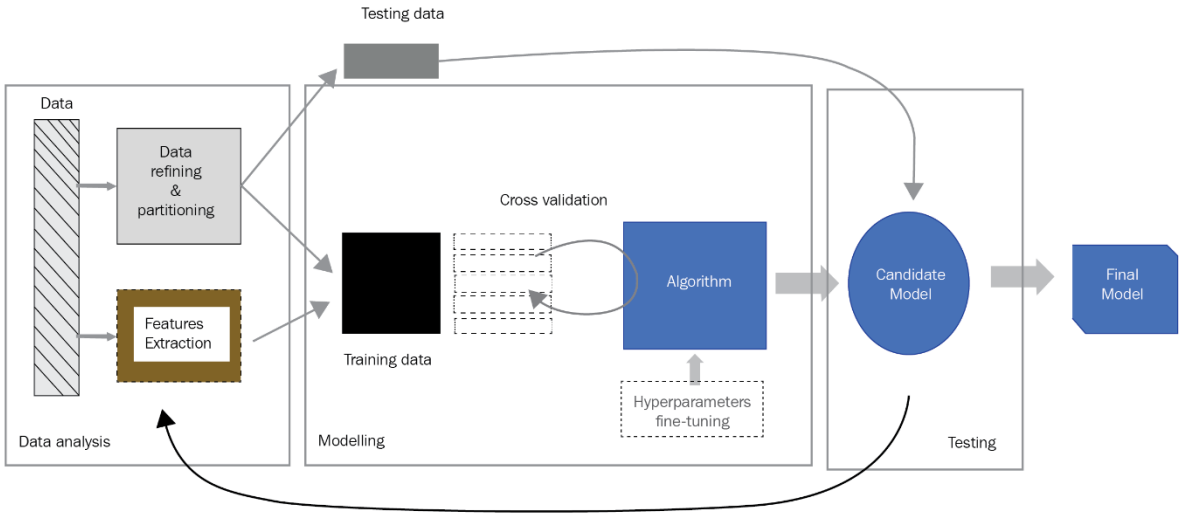
Google Storage bucket

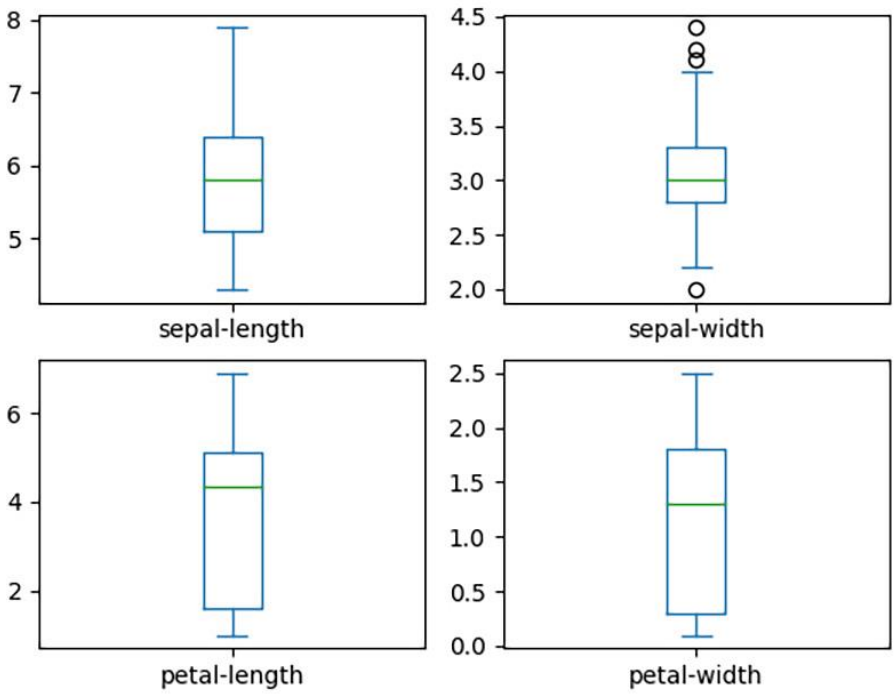
Cloud Function

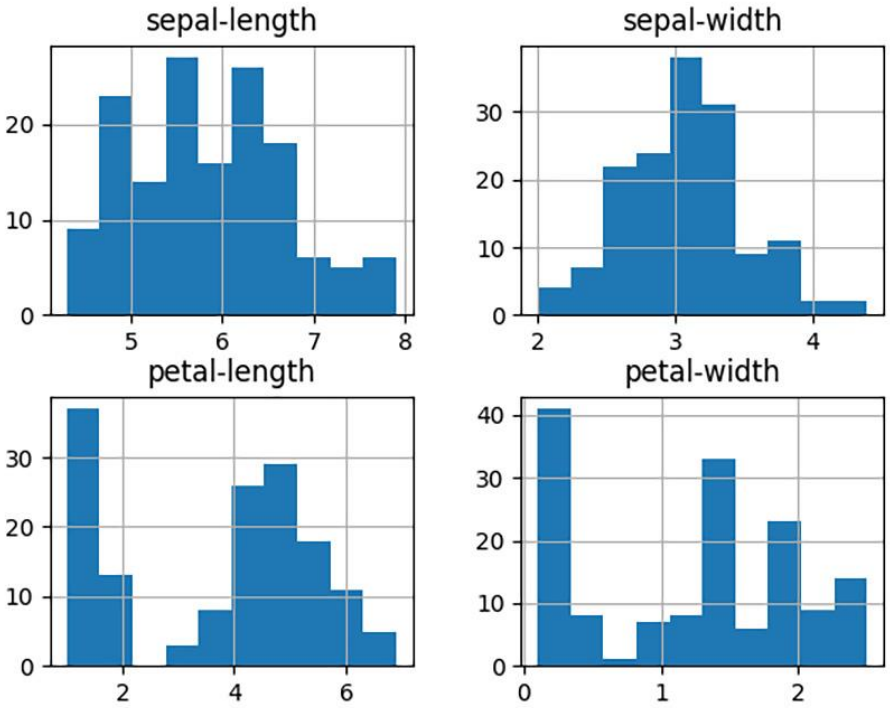
Email service

Chapter 13: Python and Machine Learning








sepal-length
sepal-width
petal-length
petal-width


k=1
k=2
k=3
k=4
k=5
Testing
Training
Training
Training
Training
Training
Testing
Training
Training
Training
Training
Training
Testing
Training
Training
Training
Training
Training
Testing
Training
Training
Training
Training
Training
Testing

Chapter 14: Using Python for Network Automation

No images…